"Calibeating" Prediction-Powered Inference

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November 11, 2026

Abstract

We study semisupervised mean estimation with a small labeled sample, a large unlabeled sample, and a black-box prediction model whose output may be miscalibrated. A standard approach in this setting is augmented inverse-probability weighting (AIPW) (Robins et al., 1994), which protects against prediction-model misspecification but may be inefficient when the prediction score is poorly aligned with the outcome scale. We introduce *Calibrated Prediction-Powered Inference*, which post-hoc calibrates the prediction score on the labeled sample before using it for semisupervised estimation. This simple step requires no retraining and can *calibeat* the original score, improving it both as a predictor of the outcome and as a regression adjustment for semisupervised inference. We study both linear and isotonic calibration. For isotonic calibration, we establish first-order optimality guarantees: isotonic post-processing can improve predictive accuracy and estimator efficiency relative to the original score and simpler post-processing rules, while replacing the fitted isotonic score by its conditional-mean oracle yields no additional first-order variance reduction. For linear calibration, we show first-order equivalence to PPI++. We also clarify the relationship between existing estimators, showing that the original PPI estimator is a special case of AIPW and can be inefficient when the prediction model is accurate, while PPI++ is AIPW with empirical efficiency maximization (Rubin and van der Laan, 2008). In simulations and real-data experiments, our calibrated estimators often outperform PPI and are competitive with, or outperform, AIPW and PPI++. We provide an accompanying `Python` package, `ppi_aipw`.

1 Introduction

We study semisupervised mean estimation with one labeled dataset $\{(X_i, Y_i)\}_{i=1}^n$ and one unlabeled dataset $\{\widetilde{X}_i\}_{i=1}^N$, where outcomes are observed only for a subset sampled completely at random. This is a classical missing-outcome problem: rich covariate information is available for many individuals, but the outcome of interest is observed only for a smaller subset. Such settings arise when gold-standard outcomes are expensive to collect but prediction scores are readily available, for example in randomized trials with prognostic models trained on historical data (Schuler et al., 2022), biobank analyses using phenotype predictors built from EHR or omics measurements (Hong et al., 2019; Zhang et al., 2019), and computational biology pipelines that supplement slow or costly experimental measurements with model-based predictions (Angelopoulos et al., 2023a; Zrnic and Candès, 2024a).

The goal is to use the larger unlabeled sample to improve estimation of a population mean while accounting for prediction error. This problem has long been studied in missing-data and semiparametric statistics, where methods such as AIPW, TMLE, and related debiased estimators combine flexible machine-learning prediction with valid inference (Robins et al., 1994; van der Laan and Robins, 2003; Van Der Laan and Rubin, 2006; van der Laan and Rose, 2011; Chernozhukov et al., 2018a). More recently, closely related ideas have reappeared in the machine-learning literature under the name prediction-powered inference (Angelopoulos

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et al., 2023a,b; Zrnic and Candès, 2024a). Similar questions also arise in covariate adjustment for randomized trials, where machine-learning predictors trained on auxiliary data can improve precision in small samples (Hansen, 2008; Rubin and van der Laan, 2008; Moore and van der Laan, 2009a; Lin, 2013; Benkeser et al., 2021; Schuler et al., 2022; Demirel et al., 2024; Jin, 2026).

A standard approach is to use the labeled sample to debias a plug-in estimator based on a prediction model. If $m(X)$ is a black-box prediction score, the plug-in estimator averages $m(X)$ over the pooled covariate sample, while the labeled sample is used to estimate the residual correction term $\mathbb{E}\{Y - m(X)\}$. For example, the augmented inverse-probability weighted (AIPW) estimator (Robins et al., 1994) averages $m(X)$ over both labeled and unlabeled samples and then adds the average residual over the labeled sample, whereas the standard PPI estimator (Angelopoulos et al., 2023a) averages $m(X)$ only over the unlabeled covariates before applying the same residual correction. Using only the unlabeled covariates can reduce efficiency when the raw score is already a good regression adjustment.

These estimators protect against bias from an imperfect prediction model, but they typically treat $m(X)$ as fixed. This can be problematic because efficiency depends on how well $m(X)$ serves as a regression adjustment, that is, on how well it approximates the conditional mean of Y, not merely on whether it ranks observations correctly. A score may be informative for ranking individuals while still being poorly calibrated, in the sense that its numerical values systematically overstate or understate the outcome. In that case, the residuals $Y - m(X)$ remain unnecessarily variable, and the resulting semisupervised estimator can be far from efficient. Such miscalibration may arise because the model was trained on a different population, or because of temporal drift, distribution shift, or regularization bias.

This motivates post-hoc calibration. Rather than correcting only the marginal bias $\mathbb{E}\{Y - m(X)\}$, one can use the labeled sample to transform $m(X)$ so that its numerical values better align with the outcomes they are meant to predict (Zadrozny and Elkan, 2001, 2002; Niculescu-Mizil and Caruana, 2005). An ideal target is a *perfectly calibrated* score $m^\star(X)$ satisfying

$$\mathbb{E}[Y \mid m^\star(X)] = m^\star(X).$$

Post-hoc calibration can improve the regression adjustment, and thus efficiency, without retraining the underlying model. Classical post-processing methods include Platt scaling (Platt et al., 1999), linear calibration (Mincer and Zarnowitz, 1969), histogram binning (Zadrozny and Elkan, 2001), and isotonic regression (Zadrozny and Elkan, 2002). Such methods are especially relevant for modern machine-learning predictors, such as neural networks, which are often informative yet systematically miscalibrated (Bella et al., 2010; Guo et al., 2017; Wang, 2023).

Contributions. We make three main contributions.

First, we introduce *calibrated prediction-powered inference* for semisupervised mean estimation (Section 3). The approach uses labeled data to calibrate an existing prediction score before using it for estimation. The resulting plug-in estimator admits an exact AIPW representation. We focus on linear and isotonic calibration, which are simple to implement, require no tuning, and do not require retraining the original predictor.

Second, we show how calibration can improve efficiency when a prediction score is informative but poorly aligned with the outcome scale. We study two related properties: improving the score as a predictor of the outcome and improving its use in semisupervised mean estimation. Following the terminology of Foster and Hart (2023), we call improvement through calibration *calibeating*. Here, these improvements concern squared prediction error and the first-order variance of semisupervised mean estimation.

Third, we establish asymptotic guarantees for linear and isotonic calibration. We show that linearly calibrated PPI is first-order equivalent to unconstrained PPI++, connecting both methods to prognostic-score regression adjustment. For isotonic calibration, we establish asymptotic normality, valid Wald inference, and efficiency guarantees (Section 4). In particular, isotonic-calibrated PPI achieves the smallest asymptotic variance among nondecreasing transformations of the original score. We further show that replacing the fitted isotonic score by its conditional-mean oracle yields no additional first-order variance reduction.

Our calibrated PPI framework can be viewed as a semisupervised specialization of the calibrated debiased

machine learning framework developed by van der Laan et al. (2024c). Our estimators can also be interpreted as targeted minimum loss estimators (Van Der Laan and Rubin, 2006; van der Laan and Rose, 2011), with calibration used in place of a conventional targeting step.

Section 2 reviews the two-sample semisupervised model, the AIPW, PPI, and PPI++ estimators, and the relevant efficiency theory. Section 5 presents simulation studies and real-data experiments. The accompanying Python package and reproduction code are publicly available at github.com/Larsvanderlaan/ppi-aipw, with examples and API documentation at larsvanderlaan.github.io/ppi-aipw/. Appendix L links to the implementations and package documentation.

1.1 Related work

PPI is closely connected to the literature on semiparametric estimation with missing outcomes, regression adjustment, and debiased machine learning. In particular, using machine-learning predictions to improve estimation efficiency is closely related to covariate adjustment in randomized trials and causal inference (Moore and van der Laan, 2009a; Benkeser et al., 2021; Højbjerre-Frandsen et al., 2025). The power-tuning approach of PPI++ can also be understood as empirical efficiency maximization (Rubin and van der Laan, 2008). Our approach builds on these connections by using calibration to improve the regression adjustment in semisupervised estimation.

Semiparametric statistics and missing data. This paper builds on foundational ideas from the missing-data and semiparametric literatures. Rubin (1976) formalized missing at random as a key identification condition for recovering full-data targets from partially observed outcomes. Semiparametric efficiency theory then characterized efficient influence functions, regular asymptotically linear estimators, and one-step debiased procedures (Levit, 1975; Pfanzagl and Wefelmeyer, 1985; Hasminskii and Ibragimov, 1979; Klaassen, 1987; Bickel et al., 1993). In missing-data and causal-inference problems, these ideas led to influence-function-based estimating equations (Robins et al., 1994, 1995; Robins and Rotnitzky, 1995; van der Laan and Robins, 2003; Tsiatis, 2006), often with doubly robust structure (Bang and Robins, 2005). As a result, the efficiency theory for missing-data models under missingness or coarsening at random is now well understood, and the class of regular asymptotically linear estimators in these models can be characterized through their influence functions (Robins et al., 1994, 1995; Robins and Rotnitzky, 1995; van der Laan and Robins, 2003; Tsiatis, 2006).

Debiased machine learning and targeted learning. A large body of work combines the semiparametric framework above with machine-learning-based nuisance estimation, often under the label of debiased machine learning. Flexible learning methods within estimating equations and AIPW-style procedures appear in early work such as van der Laan and Robins (2003). Targeted minimum loss estimation (TMLE), or targeted learning, updates an initial nuisance estimator in a parameter-targeted way to produce a plug-in estimator with reduced first-order bias; in the present setting, the primary nuisance estimated by machine learning is the outcome regression (Van Der Laan and Rubin, 2006; van der Laan and Rose, 2011; Gruber and Van Der Laan, 2009; Hoffman, 2020; Ross et al., 2025). These approaches are often combined with sample splitting and cross-fitting to accommodate generic machine-learning estimators (Schick, 1986; Klaassen, 1987; Zheng and Van Der Laan, 2010; van der Laan and Rose, 2011; Chernozhukov et al., 2018a). Double machine learning emphasizes similar ingredients—orthogonality, cross-fitting, and influence-function expansions—for inference with modern nuisance estimators (Chernozhukov et al., 2018a, 2022), and is a closely related semiparametric framework (van der Laan, 2019; Díaz, 2020; Chen et al., 2026). For sufficiently stable estimators, cross-fitting can itself be relaxed (Chen et al., 2022). Closely related work studies balancing-weight estimators (Hainmueller, 2012; Imai and Ratkovic, 2014; Zubizarreta, 2015; Lendle et al., 2015; Chattopadhyay et al., 2020; Hejazi and van der Laan, 2023), which often admit equivalent regression-adjustment or augmented representations (Chattopadhyay and Zubizarreta, 2023; Bruns-Smith et al., 2025; Rotnitzky et al., 2025). In missing-at-random settings with known missingness probability, including semisupervised learning and randomized trials, doubly robust estimators typically permit valid inference under flexible nuisance estimation even when the learned regression adjustment is misspecified (Moore and van der Laan, 2009a,b; Rosenblum and van der Laan, 2009; Højbjerre-Frandsen et al., 2025; Højbjerre-Frandsen and Schuler, 2026).

Semisupervised inference and improving power in randomized trials. Related work studies

semisupervised or surrogate-assisted estimation in settings where the primary outcome is observed only on a limited subset, while auxiliary outcomes or surrogate measurements are available more broadly (Pepe, 1992; Pepe et al., 1994; Chen, 2000; Chen et al., 2008; Cheng et al., 2021; Ji et al., 2025; Kallus and Mao, 2025). Two-sample semisupervised inference is a special case of two-stage sampling designs (Scott and Holt, 1982; Rose and van der Laan, 2011; Hejazi et al., 2021; Qiu et al., 2026) and, more generally, of data fusion, for which a substantial semiparametric efficiency literature has been developed (Li and Luedtke, 2023; Li et al., 2025; Graham et al., 2024; Xu et al., 2025). More recent papers have used the term *prediction-powered inference* (PPI) for the semisupervised mean-estimation setting in which a black-box predictor is combined with a small labeled sample and a large unlabeled sample (Angelopoulos et al., 2023a,b; Zrnic and Candès, 2024a; Xu et al., 2025; Song et al., 2026; Poulet et al., 2025; Wang et al., 2025; Lee and Kim, 2026). These papers also brought such methods to a broader machine-learning audience and provided accessible software. Closely related questions have long been studied in randomized trials, where prognostic scores or other regression adjustments learned from larger historical, external-control, or auxiliary data sources are used to improve precision in smaller trials (Hansen, 2008; Rosenblum and van der Laan, 2009; Moore and van der Laan, 2009a,b; Moore et al., 2011; VanderWeele et al., 2019; Benkeser et al., 2021; Burger et al., 2021; Schuler et al., 2022; Li et al., 2023; Balzer et al., 2024; Van Lancker et al., 2024; Højbjerre-Frandsen et al., 2025; Yang et al., 2026). Methodologically, however, PPI estimators can often be viewed as instances or refinements of existing semiparametric procedures. For example, PPI++ is an AIPW estimator (Robins et al., 1994; van der Laan and Robins, 2003) obtained by empirical efficiency maximization (Rubin and van der Laan, 2008) over a scaling parameter, while cross-PPI (Zrnic and Candès, 2024a) is a cross-fitted AIPW estimator (Zheng and Van Der Laan, 2010; Chernozhukov et al., 2018a). More fundamentally, in settings with known missingness probability, the ability to combine black-box prediction models with valid semiparametric inference is closely tied to the robustness properties of AIPW and is therefore not unique to semisupervised mean estimation (Bang and Robins, 2005; Seaman and Vansteelandt, 2018; Rotnitzky et al., 2021).

Calibrated DML and Recalibrated PPI. In our mean-estimation setting, the closest precursor is the calibrated debiased machine-learning framework of van der Laan et al. (2024c), which studies calibration of the outcome regression in AIPW estimators for doubly robust inference. Our setting can be viewed as a semisupervised specialization of that perspective, adapted to prediction-powered inference and aimed at improving efficiency in randomized designs. This connection also motivates our term *calibrated PPI*. A related method is recalibrated prediction-powered inference (RePPI) (Ji et al., 2025), which applies to general estimating equations. Rather than correcting the predictive model itself, their approach modifies the imputed loss in the PPI estimating equation, thereby "recalibrating" the estimator rather than the predictions. In the mean-estimation setting under squared loss, the target correction reduces to the conditional mean of the outcome given the initial prediction score, which is also the efficient regression adjustment in the reduced model based only on that score. Thus, this conditional mean may be viewed as an oracle calibrator. The main differences lie in the estimators, emphasis, and theory. Ji et al. (2025) use flexible machine learning to estimate this oracle target and thereby pursue semiparametric efficiency (Robins et al., 1994). They compare linear, random-forest, and neural-network recalibration, while retaining an AIPW estimator. Our focus is on simple post-hoc calibration maps applied directly to a fixed score and on guarantees that do not require consistent estimation of the oracle target. Our estimators retain a simple plug-in form, yielding lightweight methods that can be incorporated into an existing prediction pipeline without retraining. We also develop calibration-specific results: an exact AIPW interpretation of the plug-in estimator, first-order equivalence between linear calibration and PPI++, and a first-order calibeating guarantee for isotonic calibration.

From missing data to causal inference. From the perspective of estimation and efficiency theory, semisupervised mean estimation with missing-at-random outcomes is equivalent to randomized experiments, controlled trials, and survey-sampling designs once the missingness indicator is reinterpreted as a treatment or sampling indicator, so that the relevant potential outcomes are viewed as missing (Rubin, 1978; van der Laan and Robins, 2003; Ding and Li, 2018; Mozer, 2026). This equivalence allows tools from covariate adjustment, causal inference, and model-assisted estimation to be transferred directly to the semisupervised setting; see, for example, Mozer (2026) for results relating prediction-powered inference to model-assisted survey-sampling estimators. The two-sample formulation we study is also equivalent to a one-sample i.i.d. model (Li and Luedtke, 2023). In particular, our methods apply to inference on the counterfactual mean $\mathbb{E}[Y(1)]$ in a randomized trial with binary treatment, where treated units form the labeled sample and control

units contribute covariate information only; see Appendix C.

Calibration. Classical post-processing methods such as Platt scaling (Platt et al., 1999; Cox, 1958; Gupta and Ramdas, 2023), linear calibration (Mincer and Zarnowitz, 1969; Chernozhukov et al., 2018b; Leng and Dimmery, 2021), isotonic regression (Zadrozny and Elkan, 2002; Niculescu-Mizil and Caruana, 2005), and histogram binning (Zadrozny and Elkan, 2001; Gupta and Ramdas, 2021) improve predictive reliability without retraining the underlying model. The idea has deeper roots in the forecasting literature (Mincer and Zarnowitz, 1969; Lichtenstein et al., 1977; Vovk, 1992; Vovk et al., 2005; Gneiting et al., 2007; Lambert, 2011); see also the historical discussion in Lee et al. (2023). More recent work studies calibration under broader losses and prediction tasks (Jung et al., 2021; Noarov and Roth, 2023; van der Laan et al., 2023; Whitehouse et al., 2024; van der Laan and Alaa, 2025), as well as applications to conformal prediction (van der Laan and Alaa, 2024, 2025), importance-weight stabilization (Gutman et al., 2022; Deshpande and Kuleshov, 2023; Ballinari and Bearth, 2024; van der Laan et al., 2024b), treatment effect estimation (Chernozhukov et al., 2018b; van der Laan et al., 2023; Whitehouse et al., 2024), and value prediction in offline reinforcement learning (van der Laan and Kallus, 2025; van der Laan et al., 2025b). Distribution-free guarantees for histogram binning are developed by Gupta and Ramdas (2021), while van der Laan et al. (2023); van der Laan and Alaa (2025) establish asymptotic distribution-free guarantees for isotonic calibration.

2 Two-Sample Semisupervised Setup

2.1 Data structure and notation

We observe two independent datasets,

$$\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n \qquad \text{and} \qquad \mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N.$$

The labeled sample consists of i.i.d. draws from a distribution P_0 on $\mathcal{X} \times \mathbb{R}$, while the unlabeled sample consists of i.i.d. draws from the distribution $P_{0,X}$ of X under P_0. Our target is the population mean

$$\psi_0 := \mathbb{E}_{P_0}[Y], \qquad \mu_0(x) := \mathbb{E}_{P_0}[Y \mid X = x].$$

This can also be written as

$$\psi_0 = \int y \, dP_0(x, y) = \int \mu_0(x) \, dP_{0,X}(x).$$

For functions $a : \mathcal{X} \times \mathbb{R} \to \mathbb{R}$ and $b : \mathcal{X} \to \mathbb{R}$, define the empirical means over the labeled and unlabeled samples by

$$\mathbb{P}_n^L a := \frac{1}{n} \sum_{i=1}^n a(X_i, Y_i), \qquad \mathbb{P}_N^U b := \frac{1}{N} \sum_{j=1}^N b(\widetilde{X}_j).$$

For readability, we will sometimes write $\mathbb{P}_n^L\{a(X, Y)\}$ instead of $\mathbb{P}_n^L a$. When a function depends only on X, we also write

$$\mathbb{P}_n^L\{b(X)\} := \frac{1}{n} \sum_{i=1}^n b(X_i).$$

We write expectations and variances as $\mathbb{E}_{P_0}[\cdot]$, $\mathbb{E}_{P_{0,X}}[\cdot]$, $\mathrm{Var}_{P_0}[\cdot]$, and $\mathrm{Var}_{P_{0,X}}[\cdot]$.

Throughout, let $M := n + N$ and $\rho_n := n/M$ denote the labeled fraction. Then

$$\frac{1}{n+N} \left\{ \sum_{i=1}^n f(X_i) + \sum_{j=1}^N f(\widetilde{X}_j) \right\} = \rho_n \mathbb{P}_n^L\{f(X)\} + (1 - \rho_n) \mathbb{P}_N^U\{f(\widetilde{X})\}.$$

In other words, the empirical mean of $f(X)$ over the pooled covariate sample is a weighted average of the empirical means from the labeled and unlabeled samples.

Assumption 1 (Two-sample design)**.** The following hold:

(i) $\mathcal{D}_L$ is i.i.d. from P_0, and $\mathcal{D}_U$ is i.i.d. from $P_{0,X}$.

(ii) $\mathcal{D}_L$ and $\mathcal{D}_U$ are independent.

(iii) Y has finite second moment under P_0.

(iv) As $n \to \infty$, the labeled fraction $\rho_n = n/(n+N)$ satisfies $\rho_n \to \rho_0$ for some $\rho_0 \in [0, 1)$.

2.2 Review of AIPW, PPI, and semiparametric efficiency

The AIPW class. We begin with augmented inverse-probability weighting (AIPW) in the unrestricted two-sample model (Robins et al., 1994). Rather than a single estimator, AIPW defines a class indexed by a square-integrable prediction function $f : \mathcal{X} \to \mathbb{R}$. For any such f, an AIPW estimator is given by

$$\widehat{\psi}(f) := \rho_n \mathbb{P}_n^L\{f(X)\} + (1-\rho_n)\mathbb{P}_N^U\{f(\widetilde{X})\} + \mathbb{P}_n^L\{Y - f(X)\}. \tag{1}$$

The estimator uses both samples to estimate the mean of $f(X)$ and the labeled sample to estimate the mean residual. For any fixed f, $\widehat{\psi}(f)$ is unbiased in finite samples for $\psi_0 = \mathbb{E}_{P_0}[Y]$. In particular, setting $f = 0$ recovers the labeled-sample mean $\mathbb{P}_n^L\{Y\}$. The choice of f therefore affects precision but not unbiasedness. A poorly specified prediction function may forgo substantial efficiency gains, but does not introduce bias. In this sense, AIPW is a "safe" estimator (Xu et al., 2025). This robustness to the choice of f is closely related to the robustness to regression misspecification in doubly robust estimation (Bang and Robins, 2005).[1]

The central question is therefore how to choose f to minimize variance. When ρ_n is the labeled fraction of the combined sample, the variance-minimizing choice is $f_0(x) = \mathbb{E}_{P_0}[Y \mid X = x]$, up to an additive constant. Efficiency thus depends on the accuracy of conditional mean predictions. This principle also motivates the use of flexible regression methods for covariate adjustment in randomized trials (Moore and van der Laan, 2009a,b; Schuler et al., 2022; Højbjerre-Frandsen et al., 2025).

Asymptotic linearity, efficiency, and the role of residual variance. To characterize the leading large-sample error in the two-sample model, we decompose the estimation error into one contribution from the labeled sample and one from the unlabeled sample.We say that $\widehat{\psi}$ is asymptotically linear with triangular influence pair (D_n^L, D_n^U) if

$$\widehat{\psi} - \psi_0 = \frac{1}{n}\sum_{i=1}^{n} D_n^L(X_i, Y_i) + \frac{1}{N}\sum_{j=1}^{N} D_n^U(\widetilde{X}_j) + o_p(n^{-1/2}),$$

where $\mathbb{E}_{P_0}[D_n^L(X,Y)] = 0$ and $\mathbb{E}_{P_{0,X}}[D_n^U(\widetilde{X})] = 0$. If the influence functions converge in L^2 and n/N converges to a finite limit, the central limit theorem and independence of the two samples yield

$$\sqrt{n}\,(\widehat{\psi} - \psi_0) \rightsquigarrow N\Big(0,\ \lim_{n\to\infty}\Big[\mathrm{Var}_{P_0}\{D_n^L(X,Y)\} + \frac{n}{N}\mathrm{Var}_{P_{0,X}}\{D_n^U(\widetilde{X})\}\Big]\Big).$$

The influence pair thus determines the asymptotic variance, accounting for variability in both samples. The following result characterizes the influence pair for the AIPW class, identifies the efficient choice of f, and derives the semiparametric efficiency bound in the unrestricted two-sample model (Bickel et al., 1993; Robins et al., 1994; Robins and Rotnitzky, 1995).

Proposition 1 (Influence pair and efficiency of AIPW)**.** *Under Assumption 1, for any fixed square-integrable function $f : \mathcal{X} \to \mathbb{R}$, the estimator $\widehat{\psi}(f)$ is asymptotically linear with triangular influence pair $(D_{f,n}^L, D_{f,n}^U)$, where*

$$D_{f,n}^L(X,Y) = Y - \psi_0 - (1-\rho_n)\{\tilde{f}(X) - \psi_0\},$$
$$D_{f,n}^U(\widetilde{X}) = (1-\rho_n)\{\tilde{f}(\widetilde{X}) - \psi_0\},$$

[1] With unknown, covariate-dependent missingness probabilities, AIPW generally no longer enjoys finite-sample unbiasedness, but retains the usual asymptotic double robustness property (Seaman and Vansteelandt, 2018).

and $\tilde{f}(X) := f(X) - \mathbb{E}_{P_{0,X}}[f(X)] + \psi_0$ *is the recentered prediction function. The asymptotic variance of* $\sqrt{n}\{\widehat{\psi}(f) - \psi_0\}$ *is*

$$\rho_0 \mathrm{Var}_{P_0}[\mu_0(X)] + \mathbb{E}_{P_0}\big[\mathrm{Var}[Y \mid X]\big] + (1-\rho_0)\mathrm{Var}_{P_0}[f(X) - \mu_0(X)].$$

This variance is minimized when $f = \mu_0$ *up to an additive constant. Consequently, the efficient triangular influence pair in the unrestricted two-sample model is* $(D^L_{\mu_0,n}, D^U_{\mu_0,n})$*, with efficiency bound* $\rho_0 \mathrm{Var}_{P_0}[\mu_0(X)] + \mathbb{E}_{P_0}\big[\mathrm{Var}[Y \mid X]\big]$.

Proposition 1 shows that the excess asymptotic variance of AIPW relative to the efficiency bound is

$$(1-\rho_0)\mathrm{Var}_{P_0}[f(X) - \mu_0(X)].$$

Efficiency therefore depends on the accuracy of f as an approximation to the conditional mean μ_0, up to an additive constant (Wang et al., 2025). A score may rank observations correctly yet remain inefficient if its numerical values are poorly aligned with the conditional mean. This motivates the use of flexible black-box prediction scores for regression adjustment, together with post-hoc calibration to improve their approximation to the conditional mean.

Two familiar estimators arise from different choices of f in (1). Taking $f = m$, where $m(X)$ is a prediction score, yields the usual AIPW estimator (Robins et al., 1994),

$$\widehat{\psi}_{\mathrm{AIPW}} = \frac{1}{M}\left\{\sum_{i=1}^{n} m(X_i) + \sum_{j=1}^{N} m(\widetilde{X}_j) + \frac{1}{\rho_n}\sum_{i=1}^{n}\{Y_i - m(X_i)\}\right\},$$

where ρ_n^{-1} is the inverse probability of observing the outcome.

Prediction-powered inference (PPI), introduced by Angelopoulos et al. (2023a), instead uses[2]

$$\widehat{\psi}_{\mathrm{PPI}} = \mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\},$$

which corresponds to $f = m/(1-\rho_n)$ in the AIPW class. Unlike AIPW, PPI averages predictions only over the unlabeled sample, effectively rescaling the prediction score by $(1-\rho_n)^{-1} > 1$. By Proposition 1, this rescaling improves efficiency only if it reduces the variance of the prediction error relative to μ_0. In particular, when $\rho_0 > 0$ and $\mu_0(X)$ is nonconstant, PPI is inefficient even when $m = \mu_0$, whereas AIPW attains the efficiency bound. AIPW is therefore the natural default when m is intended to approximate the conditional mean (Han, 2012; Rotnitzky et al., 2012).

Empirical efficiency maximization. Both AIPW and PPI belong to the scaling class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$, corresponding to $\lambda = 1$ and $\lambda = (1-\rho_n)^{-1}$, respectively. Rather than fixing the scaling factor, one can choose it by minimizing an empirical estimate of the asymptotic variance, a procedure known as empirical efficiency maximization (Rubin and van der Laan, 2008; van der Laan and Robins, 2003). The empirical variance criterion is

$$\widehat{\sigma}_n^2(f) := \widehat{\mathrm{Var}}_L\{Y - (1-\rho_n)f(X)\} + \rho_n(1-\rho_n)\widehat{\mathrm{Var}}_U\{f(\widetilde{X})\},$$

where $\widehat{\mathrm{Var}}_L(a) := \mathbb{P}_n^L[(a - \mathbb{P}_n^L a)^2]$ and $\widehat{\mathrm{Var}}_U(b) := \mathbb{P}_N^U[(b - \mathbb{P}_N^U b)^2]$. Minimizing $\widehat{\sigma}_n^2(\lambda m)$ over $\lambda \in \mathbb{R}$ yields an AIPW estimator with the rescaled prediction function $\hat{\lambda} m$. This is equivalent, up to parameterization, to the PPI++ estimator of Angelopoulos et al. (2023b). The population-optimal scaling coefficient is

$$\lambda^\star = \frac{\mathrm{Cov}_{P_0}\{Y, m(X)\}}{\mathrm{Var}_{P_0}\{m(X)\}}, \tag{2}$$

provided $\mathrm{Var}_{P_0}\{m(X)\} > 0$. Empirical efficiency maximization also extends to richer classes of prediction functions. For example, one can minimize $\widehat{\sigma}_n^2(f)$ over

$$\left\{\lambda m(X) + \beta_{\mathrm{lin}}^\top X : \lambda \in \mathbb{R},\ \beta_{\mathrm{lin}} \in \mathbb{R}^d\right\},$$

[2]The PPI and PPI++ estimators are algebraically equivalent to the model-assisted survey-design estimators of Cassel et al. (1976) and Särndal et al. (2003), respectively; see Mozer (2026) for a review.

over monotone transformations of m, or over suitably regularized reproducing kernel Hilbert spaces; see Appendix B. Given multiple prediction scores $m_1, m_2, \ldots$, the same criterion can be used to learn an efficient linear combination. Cross-validation can further be used to select among candidate classes (Rubin and van der Laan, 2008).

3 Calibrated Prediction-Powered Inference

Standard AIPW and PPI estimators use a prediction score $m(X)$ directly for regression adjustment. Although $m(X)$ may rank observations well, its numerical values need not accurately approximate the conditional mean of the outcome. As Proposition 1 shows, such discrepancies can lead to substantial efficiency losses even when the score is highly predictive.

We propose calibrating $m(X)$ using the labeled sample before using it for semisupervised estimation. We call this approach *calibrated prediction-powered inference*. The resulting estimator averages the calibrated predictions over the pooled labeled and unlabeled samples and admits an exact AIPW representation. We study linear and isotonic calibration, which use affine and nondecreasing transformations of the original score, respectively.

3.1 Calibrated PPI and its AIPW representation

Let $m : \mathcal{X} \to \mathbb{R}$ be a black-box regression model fit using external data or a separate training sample. Given a class $\mathcal{H}$ of functions $h : \mathbb{R} \to \mathbb{R}$, we estimate a calibration map by empirical risk minimization on the labeled sample:

$$f_n \in \arg\min_{h \in \mathcal{H}} \mathbb{P}_n^L\big[(Y - h(m(X)))^2\big]. \tag{3}$$

Define the calibrated score $m_n^\star := f_n \circ m$. Our calibrated PPI estimator averages these predictions over the pooled labeled and unlabeled samples:

$$\widehat{\psi}_{\mathrm{cal}} := \frac{1}{n+N}\left\{\sum_{i=1}^{n} m_n^\star(X_i) + \sum_{j=1}^{N} m_n^\star(\widetilde{X}_j)\right\}. \tag{4}$$

Equivalently, $\widehat{\psi}_{\mathrm{cal}} = \rho_n \mathbb{P}_n^L\{m_n^\star(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n^\star(\widetilde{X})\}$.

If $\mathcal{H}$ is closed under the addition of constants, the first-order condition for (3) gives

$$\mathbb{P}_n^L\{Y - m_n^\star(X)\} = 0.$$

Consequently, the plug-in estimator (4) admits the exact AIPW representation

$$\widehat{\psi}_{\mathrm{cal}} = \rho_n \mathbb{P}_n^L\{m_n^\star(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n^\star(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m_n^\star(X)\}. \tag{5}$$

The exact AIPW representation does not imply exact unbiasedness when $m_n^\star$ is fitted using the same labeled outcomes. Because calibration uses only the labeled sample, conditioning on that sample gives, whenever the expectations are finite,

$$\mathbb{E}(\widehat{\psi}_{\mathrm{cal}} - \psi_0) = -(1-\rho_n)\mathbb{E}\big[(\mathbb{P}_n^L - P_{0,X})m_n^\star\big].$$

The right-hand side is zero for a fixed score, but need not vanish for a fitted score. The results below establish asymptotic linearity for linear and isotonic calibration without sample splitting. Section 3.4 gives exactly unbiased cross-fitted and leave-one-out AIPW estimators that retain the residual correction and exclude each observation from its calibration fit.

The goal is to obtain a calibrated score satisfying, at least approximately,

$$m_n^\star(X) = \mathbb{E}_{P_0}[Y \mid m_n^\star(X)].$$

At the population level, this property ensures that no further transformation of the calibrated score can reduce mean squared prediction error. By Proposition 1, it also ensures that no further post-processing can improve the asymptotic efficiency of the corresponding AIPW estimator. At the same time, we want to ensure that the calibrated prediction $m_n^\star(X)$ is no worse, at least asymptotically, than the original score $m(X)$ when used in the AIPW estimator.

For a fixed score q, the asymptotic variance of $\sqrt{n}\{\widehat{\psi}(q) - \psi_0\}$ can also be written as

$$V(q) = \rho_0 \mathrm{Var}(Y) + (1 - \rho_0)\mathrm{Var}\{Y - q(X)\}.$$

Adding a constant to q leaves $V(q)$ unchanged. Thus, if $\mathcal{H}$ is closed under addition of constants, the population least-squares minimizer also minimizes the asymptotic variance of AIPW over the same class of transformed scores. If the identity map belongs to $\mathcal{H}$, the resulting asymptotic variance is no greater than that of AIPW based on $m(X)$. If $\mathcal{H}$ contains the conditional mean of Y given $m(X)$, the population minimizer attains the efficiency bound in the reduced model based on $(m(X), Y)$. The choice of $\mathcal{H}$ balances flexibility to correct miscalibration against the variability of estimating the transformation from limited labeled data. We consider affine and nondecreasing transformations, corresponding to linear and isotonic calibration, respectively.

3.2 Isotonic-calibrated PPI

We consider isotonic regression as a nonparametric post-hoc calibration method (Zadrozny and Elkan, 2001, 2002; Niculescu-Mizil and Caruana, 2005). Let $\mathcal{F}_{\mathrm{iso}}$ denote the class of nondecreasing real-valued functions. We fit the calibration map using the labeled pairs $\{(m(X_i), Y_i)\}_{i=1}^n$ by solving

$$\hat{f} \in \arg\min_{f \in \mathcal{F}_{\mathrm{iso}}} \sum_{i=1}^{n} \big\{Y_i - f(m(X_i))\big\}^2,$$

and define the calibrated score

$$m_{n,\mathrm{iso}}^\star(X) = \hat{f}\{m(X)\}.$$

The resulting isotonic-calibrated PPI estimator is

$$\widehat{\psi}_{\mathrm{iso}} := \frac{1}{n+N}\left\{\sum_{i=1}^{n} m_{n,\mathrm{iso}}^\star(X_i) + \sum_{j=1}^{N} m_{n,\mathrm{iso}}^\star(\widetilde{X}_j)\right\}.$$

Isotonic regression is tuning-free and can be computed efficiently using the pooled-adjacent-violators algorithm (PAVA) (Barlow and Brunk, 1972). The fitted values on the labeled sample are constant on PAVA blocks, with each block value equal to the empirical mean outcome in that block. For new score values, we interpolate linearly between the fitted values at distinct observed scores and use the endpoint values outside the observed range. Consequently, no further transformation of the fitted score can reduce empirical squared prediction error. Thus, isotonic regression achieves full empirical calibration despite restricting the fitted transformation to be monotone. We establish its asymptotic calibration and efficiency properties in Section 4.

Proposition 2 (Isotonic calibration is fully calibrated)**.** *For every function $h : \mathbb{R} \to \mathbb{R}$,*

$$\mathbb{P}_n^L\big[h(m_{n,\mathrm{iso}}^\star(X))\{Y - m_{n,\mathrm{iso}}^\star(X)\}\big] = 0. \tag{6}$$

Moreover, for any $\theta \in \mathcal{F}_{\mathrm{iso}}$,

$$\mathbb{P}_n^L\big[(Y - m_{n,\mathrm{iso}}^\star(X))^2\big] \le \mathbb{P}_n^L\big[(Y - \theta(m(X)))^2\big].$$

In particular,

$$\mathbb{P}_n^L\big[(Y - m_{n,\mathrm{iso}}^\star(X))^2\big] \le \mathbb{P}_n^L\big[(Y - m(X))^2\big].$$

Algorithm 1 Isotonic-calibrated PPI

Require: $\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n$, $\mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N$, score m, level $1-\alpha$

1: Compute $T_i = m(X_i)$ and $\widetilde{T}_j = m(\widetilde{X}_j)$
2: Fit $\hat{f} \in \arg\min_{f \in \mathcal{F}_{\text{iso}}} \sum_{i=1}^n \{Y_i - f(T_i)\}^2$
3: Set $m^\star_{n,\text{iso}}(X_i) = \hat{f}(T_i)$ and $m^\star_{n,\text{iso}}(\widetilde{X}_j) = \hat{f}(\widetilde{T}_j)$
4: Form $\widehat{\psi}_{\text{iso}} \leftarrow (n+N)^{-1}\{\sum_{i=1}^n m^\star_{n,\text{iso}}(X_i) + \sum_{j=1}^N m^\star_{n,\text{iso}}(\widetilde{X}_j)\}$
5: Set $\widehat{D}_i^L = Y_i - \widehat{\psi}_{\text{iso}} - (1-\rho_n)\{m^\star_{n,\text{iso}}(X_i) - \widehat{\psi}_{\text{iso}}\}$ and $\widehat{D}_j^U = (1-\rho_n)\{m^\star_{n,\text{iso}}(\widetilde{X}_j) - \widehat{\psi}_{\text{iso}}\}$
6: Set $\widehat{\text{SE}}_{\text{iso}}^2 = n^{-1}\widehat{\text{Var}}_L(\widehat{D}^L) + N^{-1}\widehat{\text{Var}}_U(\widehat{D}^U)$
7: Return $\widehat{\psi}_{\text{iso}} \pm z_{1-\alpha/2}\widehat{\text{SE}}_{\text{iso}}$

Implementation. Isotonic regression is typically fitted using the pooled-adjacent-violators algorithm (PAVA) (Barlow and Brunk, 1972), which is computationally efficient but may become burdensome for very large datasets. Alternatively, monotonicity-constrained tree-based methods, such as XGBoost and LightGBM, can be used to estimate monotone calibration maps; see Appendix L. These methods also permit regularization through parameters such as the number of leaves, maximum depth, and minimum leaf size.

Histogram binning. Histogram binning partitions the score range into prespecified bins and replaces the score within each bin by the empirical mean of the outcome (Zadrozny and Elkan, 2001, 2002). Here, $\mathcal{H}$ consists of piecewise-constant functions on a fixed partition of the score space (Gupta et al., 2020; Gupta and Ramdas, 2021), and one may take $\mathcal{F} = \mathcal{H}$. Like isotonic regression, histogram binning satisfies full empirical calibration. The main difference is that histogram binning uses a fixed partition, whereas isotonic regression learns the partition from the data.

3.3 Linearly calibrated PPI

When the labeled sample is small, a parametric calibration rule can be more stable than histogram binning or isotonic regression. We therefore consider *linear calibration*, which rescales and shifts the prediction score using an affine map. This is similar in spirit to Platt scaling and temperature scaling (Platt et al., 1999; Guo et al., 2017). Unlike isotonic calibration, linear calibration cannot correct general nonlinear miscalibration, but offers a simple, low-dimensional alternative when labeled data are limited.

We estimate the calibration map by least squares on the labeled sample:

$$(\hat{a}_{\text{lin}}, \hat{b}_{\text{lin}}) \in \arg\min_{a,b \in \mathbb{R}} \sum_{i=1}^n \{Y_i - a\, m(X_i) - b\}^2, \qquad m^\star_{n,\text{lin}}(X) := \hat{a}_{\text{lin}}\, m(X) + \hat{b}_{\text{lin}}. \tag{7}$$

The resulting estimator averages the calibrated predictions over the labeled and unlabeled samples:

$$\widehat{\psi}_{\text{lin}} := \frac{1}{n+N}\left\{\sum_{i=1}^n m^\star_{n,\text{lin}}(X_i) + \sum_{j=1}^N m^\star_{n,\text{lin}}(\widetilde{X}_j)\right\}. \tag{8}$$

By construction, no further affine transformation of the fitted score can reduce empirical squared prediction error:

$$\mathbb{P}_n^L\big[(Y - m^\star_{n,\text{lin}}(X))^2\big] \le \mathbb{P}_n^L\big[(Y - a - b\, m^\star_{n,\text{lin}}(X))^2\big], \qquad a, b \in \mathbb{R}.$$

Thus, $m^\star_{n,\text{lin}}$ is empirically calibrated over the class of affine transformations. Moreover, the least-squares normal equations imply $\mathbb{P}_n^L\{Y - m^\star_{n,\text{lin}}(X)\} = 0$, so $\widehat{\psi}_{\text{lin}}$ admits the exact AIPW representation (5). The calibration regression can also include additional covariates when sufficient labeled data are available; see Algorithm 3.

The next result connects linear calibration to PPI++, which rescales the original score by empirical efficiency maximization. For mean estimation, the two methods are first-order equivalent, although they need not

Algorithm 2 Linearly calibrated PPI

Require: $\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n$, $\mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N$, score m, confidence level $1-\alpha$

1: Compute $T_i = m(X_i)$ and $\widetilde{T}_j = m(\widetilde{X}_j)$
2: Fit $(\hat{a}, \hat{b}) \in \arg\min_{a,b} \sum_{i=1}^n \{Y_i - (aT_i + b)\}^2$
3: Set $m^\star_{n,\mathrm{lin}}(X_i) = \hat{a}T_i + \hat{b}$ and $m^\star_{n,\mathrm{lin}}(\widetilde{X}_j) = \hat{a}\widetilde{T}_j + \hat{b}$
4: Compute $\widehat{\psi}_{\mathrm{lin}} \leftarrow (n+N)^{-1}\{\sum_{i=1}^n m^\star_{n,\mathrm{lin}}(X_i) + \sum_{j=1}^N m^\star_{n,\mathrm{lin}}(\widetilde{X}_j)\}$
5: Set $\widehat{D}_i^L = Y_i - \widehat{\psi}_{\mathrm{lin}} - (1-\rho_n)\{m^\star_{n,\mathrm{lin}}(X_i) - \widehat{\psi}_{\mathrm{lin}}\}$ and $\widehat{D}_j^U = (1-\rho_n)\{m^\star_{n,\mathrm{lin}}(\widetilde{X}_j) - \widehat{\psi}_{\mathrm{lin}}\}$
6: Compute $\widehat{\mathrm{SE}}^2_{\mathrm{lin}} = n^{-1}\widehat{\mathrm{Var}}_L(\widehat{D}^L) + N^{-1}\widehat{\mathrm{Var}}_U(\widehat{D}^U)$
7: **return** $\widehat{\psi}_{\mathrm{lin}} \pm z_{1-\alpha/2}\widehat{\mathrm{SE}}_{\mathrm{lin}}$

Algorithm 3 Linearly calibrated PPI with covariate adjustment

Require: $\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n$, $\mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N$, score m, confidence level $1-\alpha$

1: Compute $T_i = m(X_i)$ and $\widetilde{T}_j = m(\widetilde{X}_j)$
2: Fit $(\hat{a}, \hat{\beta}, \hat{b}) \in \arg\min_{a,\beta,b} \sum_{i=1}^n \{Y_i - (aT_i + X_i^\top\beta + b)\}^2$
3: Set $m^\star_{n,\mathrm{lin}}(X_i) = \hat{a}T_i + X_i^\top\hat{\beta} + \hat{b}$ and $m^\star_{n,\mathrm{lin}}(\widetilde{X}_j) = \hat{a}\widetilde{T}_j + \widetilde{X}_j^\top\hat{\beta} + \hat{b}$
4: Compute $\widehat{\psi}_{\mathrm{lin}} \leftarrow (n+N)^{-1}\{\sum_{i=1}^n m^\star_{n,\mathrm{lin}}(X_i) + \sum_{j=1}^N m^\star_{n,\mathrm{lin}}(\widetilde{X}_j)\}$
5: Set $\widehat{D}_i^L = Y_i - \widehat{\psi}_{\mathrm{lin}} - (1-\rho_n)\{m^\star_{n,\mathrm{lin}}(X_i) - \widehat{\psi}_{\mathrm{lin}}\}$ and $\widehat{D}_j^U = (1-\rho_n)\{m^\star_{n,\mathrm{lin}}(\widetilde{X}_j) - \widehat{\psi}_{\mathrm{lin}}\}$
6: Compute $\widehat{\mathrm{SE}}^2_{\mathrm{lin}} = n^{-1}\widehat{\mathrm{Var}}_L(\widehat{D}^L) + N^{-1}\widehat{\mathrm{Var}}_U(\widehat{D}^U)$
7: **return** $\widehat{\psi}_{\mathrm{lin}} \pm z_{1-\alpha/2}\widehat{\mathrm{SE}}_{\mathrm{lin}}$

coincide in finite samples. Both target the population-optimal scaling coefficient $\lambda^\star$ in (2), corresponding to the slope in the population linear regression of Y on $m(X)$. This equivalence requires optimization over all $\lambda \in \mathbb{R}$. Restricting λ to $[0, (1-\rho_n)^{-1}]$, as in the official `ppi_py` implementation, can break this equivalence and reduce efficiency, as illustrated in our experiments.

Let $\widehat{\psi}_{++}(\widehat{\lambda}_{++})$ denote the unconstrained PPI++ estimator of Angelopoulos et al. (2023b), namely, the AIPW estimator based on the rescaled score $\widehat{\lambda}_{++}m(X)$, where $\widehat{\lambda}_{++}$ is chosen by empirical efficiency maximization over $\{\lambda m(X) : \lambda \in \mathbb{R}\}$ (Rubin and van der Laan, 2008).

Proposition 3 (First-order equivalence of PPI++ and linear calibration)**.** *Under Assumption 1, suppose that* $\mathrm{Var}[m(X)] > 0$ *and* $\mathbb{E}_{P_0}[m(X)^2] < \infty$*. Then*

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\mathrm{lin}} = (1-\rho_n)\big(\hat{\lambda}_{++} - \hat{a}_{\mathrm{lin}}\big)\left\{\mathbb{P}_N^U\{m(\widetilde{X})\} - \mathbb{P}_n^L\{m(X)\}\right\}.$$

Moreover, $\hat{\lambda}_{++} - \hat{a}_{\mathrm{lin}} = o_p(1)$*, and*

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\mathrm{lin}} = o_p(n^{-1/2}).$$

Remark 1 (Relation to prognostic-score adjustment)**.** Linear calibration can be viewed as prognostic-score regression adjustment, a classical approach to improving precision through covariate adjustment (Hansen, 2008; Rosenblum and van der Laan, 2009; Lin, 2013; Schuler et al., 2022; Balzer et al., 2024; Højbjerre-Frandsen et al., 2025). Proposition 3 shows that PPI++ has the same first-order interpretation. Linear calibration can also be viewed as a special case of TMLE (Højbjerre-Frandsen and Schuler, 2026).

Remark 2 (Platt-scaling adjustment)**.** Linear calibration extends beyond squared-error loss to logistic, Poisson, and other losses associated with canonical-link generalized linear models (Rosenblum and van der Laan, 2009). For binary outcomes $Y \in \{0,1\}$ and an initial prediction score $m(X) \in (0,1)$, one can use Platt scaling (Cox, 1958; Platt et al., 1999), which fits a logistic regression of Y on the logit-transformed score:

$$\big(\hat{a}_{\mathrm{platt}}, \hat{b}_{\mathrm{platt}}\big) \in \arg\min_{a,b\in\mathbb{R}} \sum_{i=1}^n \Big[-Y_i \log\sigma\big(a\,\mathrm{logit}(m(X_i)) + b\big) - (1-Y_i)\log\Big(1 - \sigma\big(a\,\mathrm{logit}(m(X_i)) + b\big)\Big)\Big], \quad (9)$$

where $\sigma(t) := (1 + e^{-t})^{-1}$. The calibrated predictor is

$$m^{\star}_{n,\mathrm{platt}}(X) := \sigma\big(\hat{a}_{\mathrm{platt}} \operatorname{logit}(m(X)) + \hat{b}_{\mathrm{platt}}\big).$$

Platt scaling can be useful when predicted probabilities are systematically over- or under-confident and the calibration curve is approximately sigmoidal. More generally, outcomes can be rescaled to $[0, 1]$ before applying this procedure. Calibration properties of Platt scaling in online settings are studied by Gupta and Ramdas (2023).

3.4 Cross-fitting and leave-one-out calibration

Fitting a calibrator to pooled out-of-fold predictions still uses each labeled outcome in the calibration fit. To hold out both steps, partition the labeled indices into folds $I_1, \ldots, I_K$ independently of the observations, and fit each calibrated score $q^{(-k)}$ without the pairs indexed by I_k. Any predictor fitting and tuning used to construct $q^{(-k)}$ also exclude I_k. Let $k(i)$ be the fold containing i and $c_n := N/(n+N)$. The cross-fitted AIPW estimator is

$$\widehat{\psi}_{\mathrm{CF}} := \frac{1}{n}\sum_{i=1}^{n}\left[Y_i + c_n\{\mathbb{P}_N^U q^{(-k(i))} - q^{(-k(i))}(X_i)\}\right]. \tag{10}$$

For each i, conditional on the training sample, $\mathbb{E}[q^{(-k(i))}(X_i)] = \mathbb{E}[\mathbb{P}_N^U q^{(-k(i))}]$. Thus (10) is unbiased when these predictions are integrable and the fitted scores do not use the unlabeled observations.

Leave-one-out calibration takes $K = n$. Writing q_{-i} for the score fitted without (X_i, Y_i), we obtain

$$\widehat{\psi}_{\mathrm{LOO}} = \frac{1}{n}\sum_{i=1}^{n}\left[Y_i + c_n\{\mathbb{P}_N^U q_{-i} - q_{-i}(X_i)\}\right]. \tag{11}$$

The residual correction is retained because the held-out residuals need not average to zero. In the simulation, we use

$$\widehat{\mathrm{se}}^2_{\mathrm{LOO}} = \frac{\widehat{\mathrm{Var}}_L\{Y_i - c_n q_{-i}(X_i)\}}{n} + \frac{\widehat{\mathrm{Var}}_U\{c_n \bar{q}(\widetilde{X}_j)\}}{N}, \qquad \bar{q} := \frac{1}{n}\sum_{i=1}^{n} q_{-i}, \tag{12}$$

where the empirical variances use denominators $n-1$ and $N-1$. The bias, coverage, and interval widths of this procedure are reported in Appendix K.1.

4 Theory for isotonic-calibrated PPI

We study the theoretical properties of isotonic-calibrated PPI, showing how calibration can improve both prediction accuracy and the efficiency of semisupervised mean estimation. We establish asymptotic linearity, asymptotic normality, and valid Wald inference for $\widehat{\psi}_{\mathrm{iso}}$. We then compare its asymptotic variance with that of AIPW, PPI, PPI++, and linear calibration, and give conditions under which it attains the semiparametric efficiency bound based on either $m(X)$ or the full covariate vector X (Bickel et al., 1993; Van der Vaart, 2000). Finally, we show that the fitted isotonic score is asymptotically calibrated at the population level. Replacing the fitted isotonic score by its conditional-mean oracle yields no additional first-order variance reduction.

4.1 Asymptotic normality, inference, and efficiency

For a fixed initial score m, define the population isotonic calibration map

$$f_0 \in \arg\min_{f \in \mathcal{F}_{\mathrm{iso}}} \mathbb{E}_{P_0}\big[\{Y - f(m(X))\}^2\big], \qquad m_0 = f_0 \circ m. \tag{13}$$

Thus, m_0 is the population L^2-optimal monotone transformation of $m(X)$. Since the identity map belongs to $\mathcal{F}_{\mathrm{iso}}$, its mean squared prediction error is no greater than that of the original score. Under the following

regularity conditions, the isotonic estimator $m^\star_{n,\mathrm{iso}}$ achieves the classical $n^{-2/3}$ mean squared error rate (Chatterjee et al., 2015; Yang and Barber, 2018).

Assumption 2 (Regularity conditions for isotonic calibration)**.** Assume:

(i) **Boundedness:** There exists $C_0 < \infty$ such that $|\mathbb{E}_{P_0}[Y \mid m(X)]| \le C_0$ almost surely and $\sup_x |m^\star_{n,\mathrm{iso}}(x)| = O_p(1)$.

(ii) **Controlled tails:** For $T = m(X)$, there exists $K < \infty$ such that

$$\mathbb{E}_{P_0}\left[\exp\left\{\frac{|Y - \mathbb{E}_{P_0}[Y \mid T]|}{K}\right\}\middle| T\right] \le 2 \qquad \text{almost surely.}$$

Condition (i) requires a bounded conditional mean and a fitted isotonic regression that remains bounded in probability. Condition (ii) permits unbounded outcomes with sub-Gaussian or sub-exponential errors, covering binary, Gaussian, Poisson, and bounded outcomes. Results for isotonic regression under heavier-tailed errors are also available (Han and Wellner, 2018, 2019).

Theorem 1 (Asymptotic linearity under the two-sample model)**.** *Suppose Assumptions 1 and 2 hold. Then*

$$\|m^\star_{n,\mathrm{iso}} - m_0\|^2_{2,P_{0,X}} = O_p(n^{-2/3}).$$

Define

$$D^L_{m_0,n}(X,Y) := Y - \psi_0 - (1-\rho_n)\{m_0(X) - \psi_0\}, \qquad D^U_{m_0,n}(\widetilde{X}) := (1-\rho_n)\{m_0(\widetilde{X}) - \psi_0\}.$$

Then

$$\begin{aligned} \widehat{\psi}_{\mathrm{iso}} - \psi_0 &= (\mathbb{P}^L_n - P_0)D^L_{m_0,n} + (\mathbb{P}^U_N - P_{0,X})D^U_{m_0,n} + R_{n,N}, \\ R_{n,N} &= O_p\big((1-\rho_n)n^{-2/3}\big) + O_p\big((1-\rho_n)n^{-1/3}N^{-1/2}\big) = o_p(n^{-1/2}). \end{aligned} \tag{14}$$

Consequently, $\widehat{\psi}_{\mathrm{iso}}$ is asymptotically linear at the $\sqrt{n}$ rate with triangular influence pair $(D^L_{m_0,n}, D^U_{m_0,n})$.

The asymptotic linear representation yields a mean-zero normal limit for $\sqrt{n}(\widehat{\psi}_{\mathrm{iso}} - \psi_0)$ with variance

$$\sigma_0^2 := \mathrm{Var}[Y - \psi_0 - (1-\rho_0)\{m_0(X) - \psi_0\}] + \rho_0(1-\rho_0)\mathrm{Var}\{m_0(X)\}.$$

The following corollary establishes asymptotic normality and valid Wald inference using an empirical estimate of this variance.

Corollary 1 (Asymptotic normality and Wald inference)**.** *Under the conditions of theorem 1, assume that $\sigma_0^2 > 0$. Define the empirical variance estimator*

$$\widehat{\sigma}^2 := \mathbb{P}^L_n\{(\widehat{D}^L - \mathbb{P}^L_n\widehat{D}^L)^2\} + \frac{n}{N}\mathbb{P}^U_N\{(\widehat{D}^U - \mathbb{P}^U_N\widehat{D}^U)^2\},$$

where

$$\widehat{D}^L_i := Y_i - \widehat{\psi}_{\mathrm{iso}} - (1-\rho_n)\{m^\star_{n,\mathrm{iso}}(X_i) - \widehat{\psi}_{\mathrm{iso}}\}, \qquad \widehat{D}^U_j := (1-\rho_n)\{m^\star_{n,\mathrm{iso}}(\widetilde{X}_j) - \widehat{\psi}_{\mathrm{iso}}\}.$$

Then

$$\frac{\sqrt{n}\,(\widehat{\psi}_{\mathrm{iso}} - \psi_0)}{\widehat{\sigma}} \xrightarrow{d} N(0,1).$$

We next characterize the efficiency of isotonic-calibrated PPI relative to other transformations of the initial score. The distinction is between efficiency in the full two-sample model and efficiency when regression adjustment is restricted to functions of $m(X)$.

Corollary 2 (Efficiency regimes)**.** *Under the conditions of theorem 1, the following hold.*

(i) If $m_0 = \mu_0$ almost surely, then $\widehat{\psi}_{\text{iso}}$ is semiparametrically efficient for ψ_0 in the full two-sample model.

(ii) More generally, $\widehat{\psi}_{\text{iso}}$ is asymptotically at least as efficient as any AIPW estimator of the form $\widehat{\psi}(\theta \circ m)$, where θ is nondecreasing and $\theta \circ m$ is square-integrable.

(iii) If $\mathbb{E}_{P_0}[Y \mid m(X)]$ is monotone nondecreasing in $m(X)$, then $m_0(X) = \mathbb{E}_{P_0}[Y \mid m(X)]$ almost surely. In this case, no AIPW estimator based on a square-integrable transformation $\theta \circ m$, with $\theta : \mathbb{R} \to \mathbb{R}$, has smaller asymptotic variance than $\widehat{\psi}_{\text{iso}}$. Equivalently, $\widehat{\psi}_{\text{iso}}$ is semiparametrically efficient in the reduced two-sample model obtained by replacing X with $m(X)$.

Taking θ constant in part (ii) shows that isotonic calibration is asymptotically at least as efficient as the labeled mean. At the $\sqrt{n}$ scale, the variance reduction is

$$\text{Var}(Y) - \sigma_0^2 = (1 - \rho_0)\text{Var}\{m_0(X)\}.$$

Corollary 2 shows that isotonic calibration is at least as efficient as other monotone transformations, including AIPW, PPI, PPI++, and linear calibration when the corresponding scaling or slope coefficient is nonnegative. The improvement can be strict when the optimal transformation is nonlinear. If the score is negatively associated with the outcome, its sign can be reversed before isotonic calibration. Full semiparametric efficiency is attained when the calibrated score recovers the conditional mean $\mu_0(X)$. Inefficiency can arise either because the scalar score loses information about the conditional mean or because the monotonicity constraint prevents its recovery from the score.

When $N/n \to \infty$, the limiting labeled fraction is zero, and the asymptotic variance reduces to

$$\sigma_0^2 = \text{Var}\{Y - m_0(X)\}.$$

If the conditional mean of Y given $m(X)$ is nondecreasing in the score, this equals $\mathbb{E}[\text{Var}\{Y \mid m(X)\}]$. The unlabeled contribution is negligible at the $\sqrt{n}$ scale, and the labeled residuals determine first-order uncertainty.

4.2 First-order equivalence to oracle calibration

We next consider whether further post-processing of the isotonically calibrated score can improve prediction or estimation efficiency. Since $\widehat{\psi}_{\text{iso}}$ depends on the covariates only through the fitted score $m^\star_{n,\text{iso}}(X)$, we consider the reduced two-sample experiment

$$\left(\{(m^\star_{n,\text{iso}}(X_i), Y_i)\}_{i=1}^n,\ \{m^\star_{n,\text{iso}}(\widetilde{X}_j)\}_{j=1}^N\right),$$

in which the full covariate vector X is replaced by $m^\star_{n,\text{iso}}(X)$.

The optimal regression adjustment in this reduced experiment is $\mathbb{E}_{P_0}[Y \mid m^\star_{n,\text{iso}}(X)]$, where the conditional expectation is evaluated for a new observation from P_0, treating the data used to fit $m^\star_{n,\text{iso}}$ as fixed. We show that replacing the isotonic score by this oracle regression yields no first-order improvement in estimation efficiency.

Assumption 3 (Bounded-variation score regression)**.** The map $t \mapsto \mathbb{E}_{P_0}[Y \mid m(X) = t]$ has bounded variation on the support of $m(X)$.

Assumption 3 requires the conditional mean of Y given the initial score to have bounded total variation. It permits nonmonotone functions and discontinuities, and does not require differentiability or Lipschitz continuity (van der Laan et al., 2023).

Theorem 2 (First-order equivalence to oracle calibration)**.** *Suppose Assumptions 1 to 3 hold. Let*

$$\widehat{\psi}_{\text{eff}} := \widehat{\psi}(m_{0|n}), \qquad m_{0|n}(X) := \mathbb{E}_{P_0}[Y \mid m^\star_{n,\text{iso}}(X)]$$

denote the oracle AIPW estimator in the reduced experiment based on the fitted isotonic score. Then

$$\|m_{0|n} - m^{\star}_{n,\mathrm{iso}}\|^2_{2,P_{0,X}} = O_p(n^{-2/3}),$$

and

$$\widehat{\psi}_{\mathrm{eff}} - \widehat{\psi}_{\mathrm{iso}} = o_p(n^{-1/2}).$$

In particular, $\widehat{\psi}_{\mathrm{iso}}$ *and* $\widehat{\psi}_{\mathrm{eff}}$ *have the same asymptotic variance.*

Theorem 2 establishes that the fitted isotonic score is calibrated up to an $O_p(n^{-2/3})$ squared L^2 error. Together with the convergence result in theorem 1 and the population optimality of m_0, this shows that isotonic calibration does not asymptotically increase mean squared prediction error relative to the original score. This is calibeating in terms of squared prediction error.

The theorem also establishes first-order equivalence between $\widehat{\psi}_{\mathrm{iso}}$ and the oracle AIPW estimator based on $m_{0|n}$. Thus, replacing the fitted isotonic score by its conditional-mean oracle yields no additional first-order variance reduction. Moreover, by theorem 1 and proposition 1, isotonic calibration achieves the smallest limiting variance among monotone post-processings of the original score and is therefore at least as efficient as AIPW and PPI. This is calibeating in terms of first-order estimation efficiency.

5 Experiments

We compare the proposed calibration methods against the labeled-only estimator, standard AIPW and PPI, and the empirical efficiency maximization methods PPI++ and AIPW-EM. The calibration methods include `LinearCal`, `IsoCal`, and `AutoCal`. We first examine the effects of score miscalibration in controlled simulations, then evaluate the methods on established benchmarks and an application involving LLM-generated predictions. Empirical relative efficiency is the ratio of the reference estimator's Monte Carlo variance to that of the estimator being evaluated; each caption specifies the reference estimator.

5.1 Simulation study

We consider a binary-outcome simulation designed to examine the effects of score miscalibration. Let $S \sim N(0,1)$, define

$$\mu_0(S) = \{1 + \exp(-5S)\}^{-1},$$

and generate $Y \mid S \sim \mathrm{Bernoulli}\{\mu_0(S)\}$. We construct a miscalibrated prediction score by applying a nonlinear monotone transformation:

$$m(X) = \mathrm{clip}\left[-0.15 + 0.75\,\sigma\big\{3\,z(S) - 0.5\big\},\ 0.01,\ 0.99\right],$$

where $z(S) := \mathrm{logit}\{\mu_0(S)\}$ and $\sigma(t) := \{1 + \exp(-t)\}^{-1}$. This transformation preserves the ordering of the conditional mean, apart from ties induced by clipping, while introducing substantial miscalibration on the probability scale.

We compare the labeled-only mean (Labeled-only), PPI, AIPW, PPI++, empirical efficiency-maximized AIPW (AIPW-EM), linear calibration (LinearCal), monotone spline calibration (MonoSpline) (Ramsay, 1988; Jiang et al., 2011), isotonic calibration (IsoCal), and an adaptive procedure (AutoCal) that selects among AIPW and the calibration methods by cross-validated empirical efficiency. For PPI++, we use the official `ppi_py` implementation, which selects a scaling coefficient by empirical efficiency maximization and restricts it to $[0,1]$ in the original PPI parameterization, equivalently $[0,(1-\rho_n)^{-1}]$ in the AIPW parameterization. By contrast, AIPW-EM minimizes the same criterion without restricting the coefficient. We vary the labeled sample size over $n \in \{50, 100, 200, 400, 800, 1200, 2400\}$ and consider two unlabeled-to-labeled sample size ratios, $N/n \in \{1, 16\}$. Results are based on 500 Monte Carlo repetitions per configuration.

Figure 1 shows that calibration improves efficiency when the prediction score is miscalibrated. At $n = 400$ and $N/n = 16$, `LinearCal`, `MonoSpline`, and `IsoCal` have RMSEs between 0.0164 and 0.0166, compared

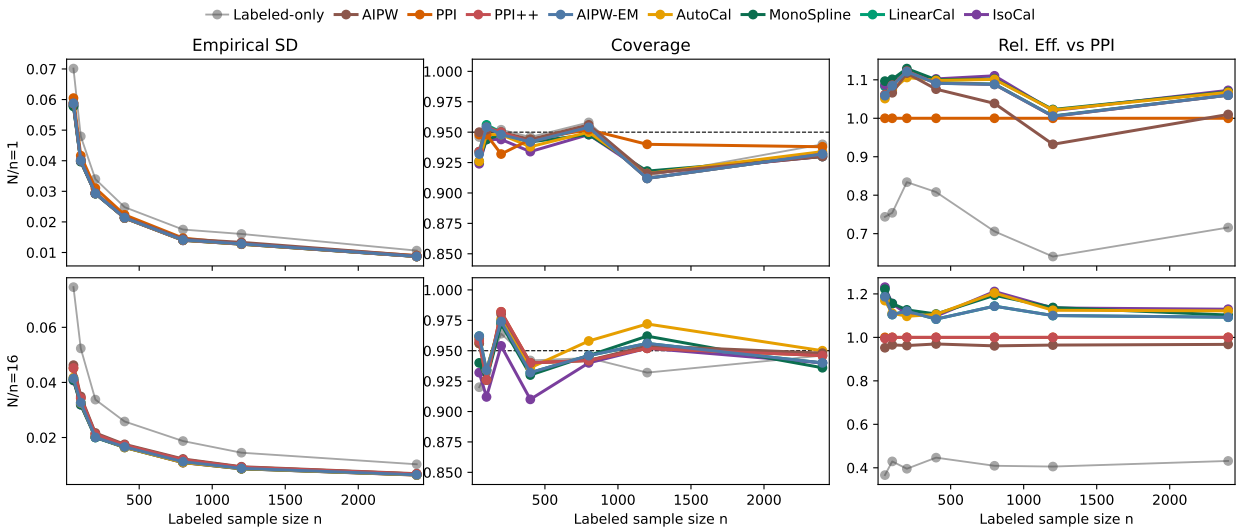


Figure 1: Simulation results with a miscalibrated prediction score. The top and bottom rows correspond to $N/n = 1$ and $N/n = 16$, respectively. The columns show Monte Carlo standard deviation, coverage of nominal 95% confidence intervals, and relative efficiency compared with PPI, with values above one indicating greater efficiency. Bias was negligible relative to standard errors and is omitted.

with 0.0172 for PPI and 0.0175 for AIPW. Their nominal 95% Wald-interval coverage ranges from 0.910 to 0.932. At $n = 1200$, AutoCal, MonoSpline, and IsoCal achieve RMSEs of approximately 0.0087, compared with 0.0092 for PPI.

The comparison between PPI++ and AIPW-EM illustrates the effect of restricting the scaling coefficient. AIPW-EM performs similarly to LinearCal, whereas PPI++ approaches the performance of unadjusted PPI when the unlabeled sample is large. Averaged across sample sizes with $N/n = 16$, RMSE is approximately 0.0196 for AIPW-EM and LinearCal, compared with 0.0208 for PPI++. This difference is consistent with the constraint imposed by the `ppi_py` implementation, which restricts the PPI++ scaling coefficient to $[0, 1]$, equivalently $[0, (1 - \rho_n)^{-1}]$ in the AIPW parameterization. Finally, AutoCal performs comparably to the best fixed calibration methods, while MonoSpline achieves similar accuracy to IsoCal.

5.2 Small-sample comparisons

We also examine smaller labeled samples under the same binary-outcome model, using mild and strong score miscalibration as specified in Appendix K.1. We set $N = 16n$ for $n \in \{20, 50, 100, 200, 400, 800, 1200\}$ and use 1,000 replications per setting. The labeled mean, PPI, AIPW, PPI++, linear, smooth monotone, and isotonic calibration are evaluated on the same samples, together with the leave-one-out linear and isotonic estimators in (11).

Under strong miscalibration, linear, smooth monotone, and isotonic calibration each have lower RMSE than PPI++ at all seven sample sizes. At $n = 20$, isotonic coverage ranges from 0.814 to 0.871 across the two settings; leave-one-out calibration increases it to 0.905–0.925, with wider intervals and larger RMSE. Across all 14 settings, leave-one-out calibration lowers linear-calibration RMSE by at most 1.99%, but lowers isotonic RMSE in only one setting. The RePPI random-forest and neural-network comparisons are reported in Appendix K.1.

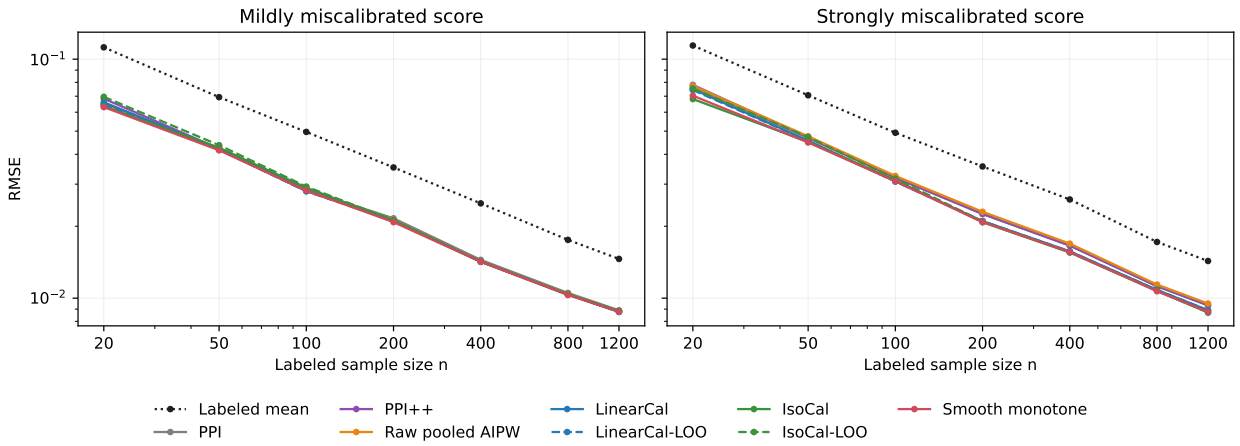


Figure 2: RMSE across labeled sample sizes under mild and strong score miscalibration, with $N = 16n$ and 1,000 replications per setting. Solid and dashed curves distinguish calibration on the labeled sample from leave-one-out calibration.

5.3 Empirical benchmarks

We evaluate the proposed estimators on the `forest`, `galaxies`, and `census_income` benchmarks used by Angelopoulos et al. (2023a,b) and distributed with the `ppi_py` package. Following the original labeled-sample-size grids, we repeatedly partition each fully labeled dataset into labeled and unlabeled samples, evaluate all estimators on the same split, and use the full-sample mean as the reference value. Results are based on 500 random splits per configuration.[3] We compare PPI, AIPW, PPI++, AIPW-EM, `LinearCal`, and `AutoCal`. Additional calibration methods and dataset-specific results are reported in Appendix K. In these benchmarks and the LLM experiments, `LinearCal` clips affine predictions to the labeled outcome range, and `IsoCal` first groups observations into bins of at least 10 before fitting isotonic regression.

Figure 3 shows that AIPW generally improves on PPI, particularly when the unlabeled sample is not much larger than the labeled sample. Calibration provides further gains, most notably on `census_income`, where `LinearCal` achieves the lowest average MSE. The isotonic and monotone spline methods perform similarly to the leading estimators (Figure 7).

The comparison with PPI++ also shows the effect of restricting the empirical efficiency-maximizing coefficient. Across the 30 dataset-size configurations, AIPW-EM achieves lower MSE than PPI++ in 29 and performs nearly identically to `LinearCal`. All three methods have similar empirical coverage, averaging approximately 92% across configurations with nominal coverage of either 90% or 95%. Thus, much of the improvement over PPI++ is attributable to removing the constraint on its scaling coefficient, rather than to nonlinear calibration. `AutoCal` also performs comparably to the best fixed method across the benchmarks.

5.4 LLM evaluation benchmark

We evaluate the methods on two public LLM benchmarks. The first is PPE Human Preference V1 (Frick et al., 2025), where the estimand for each target model is its human win rate across prompt–response comparisons, counting ties as half-wins. The second is a PPE correctness benchmark (Frick et al., 2025) constructed from MMLU-Pro (Wang et al., 2024), MATH (Hendrycks et al., 2021), GPQA (Rein et al., 2024), IFEval (Zhou et al., 2023), and MBPP+ (Liu et al., 2023). Each observation consists of a sampled pair containing one correct and one incorrect response from the same model, and the estimand is the probability that the first response is correct. We use the public model outputs, labels, and evaluator scores, randomizing only the

[3]Coverage is evaluated over repeated partitions of a fixed dataset, rather than independent draws from a superpopulation. Consequently, coverage of Wald confidence intervals need not equal the nominal level.

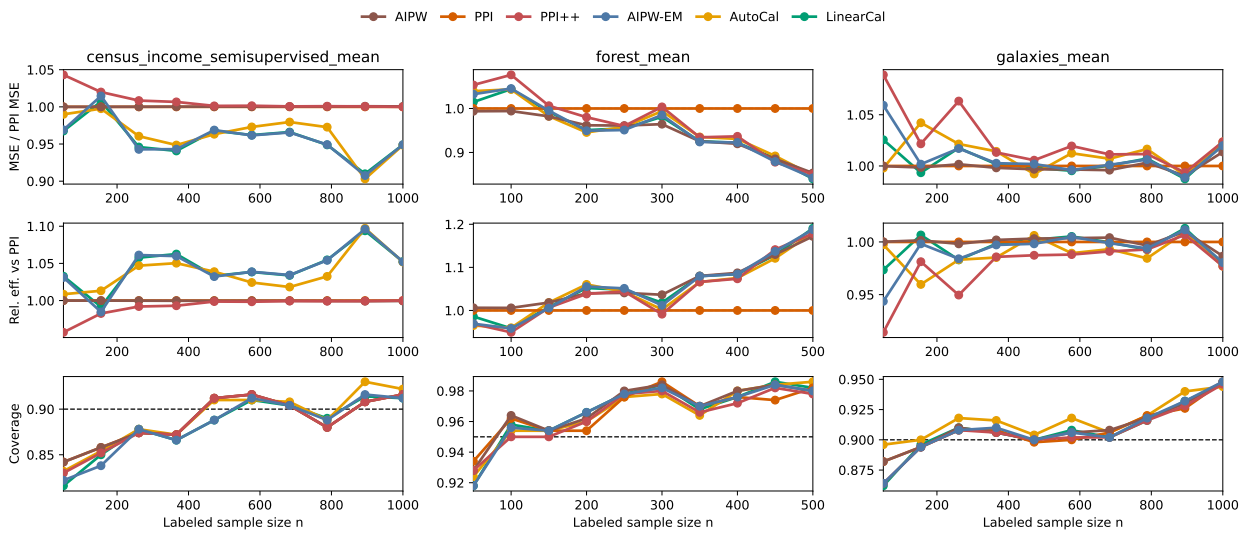


Figure 3: Results on the `ppi_py` benchmarks. Panels show MSE normalized by that of PPI, relative efficiency compared with PPI, and confidence-interval coverage. Nominal coverage is 95% for `forest` and 90% for `galaxies` and `census_income`.

labeled and unlabeled split. Results are based on 200 repeated splits. We compare labeled-only estimation, PPI, AIPW, PPI++, AIPW-EM, and the proposed calibration methods.

For each benchmark, evaluator, and target model, MSE/PPI is the estimator's Monte Carlo MSE divided by that of PPI. Implied label savings are

$$\max\left\{0,\, 1 - \frac{\widehat{\mathrm{Var}}(\widehat{\psi})}{\widehat{\mathrm{Var}}(\overline{Y}_L)}\right\},$$

where the variances are computed over repeated splits and $\overline{Y}_L$ is the labeled mean. We set negative savings to zero before averaging. Table entries are averaged first over target models within each benchmark, then over benchmarks within each evaluator, and then over evaluators.

Raw evaluator scores can substantially increase estimation variance. On PPE Human at $n = 100$, PPI and AIPW have MSEs of 0.495 and 0.440, respectively, compared with 0.001425 for labeled-only estimation. The corresponding MSEs on PPE Correctness are 0.534, 0.353, and 0.002010. Although AIPW improves on PPI, both perform considerably worse than ignoring the evaluator scores. The largest losses are associated with evaluators such as Skywork and Athene, whereas ArmoRM exhibits substantially less miscalibration (Figure 10).

Calibration largely eliminates these efficiency losses. On PPE Human, `LinearCal` has MSE 0.001391 at $n = 100$ and 0.000278 at $n = 400$, compared with 0.001425 and 0.000286 for labeled-only estimation. Relative to labeled-only estimation, these are MSE reductions of 2.4% and 2.6%, respectively. The gains are larger on PPE Correctness, where `LinearCal` reduces MSE from 0.002010 to 0.001813 at $n = 100$ and from 0.000248 to 0.000222 at $n = 400$, corresponding to MSE reductions of 9.8% and 10.5%, respectively. At $n = 100$, its nominal 90% Wald intervals have empirical coverage of 0.909 on PPE Human and 0.931 on PPE Correctness. `AutoCal` performs similarly, while MonoSpline provides gains over raw PPI and AIPW and remains competitive with PPI++ and AIPW-EM. The relatively strong performance of linear calibration suggests that correcting the scale of the evaluator scores accounts for much of the available efficiency gain.

Calibration also improves model ranking on PPE Human. Raw PPI and AIPW recover the model ordering poorly, whereas calibrated and efficiency-maximized estimators perform substantially better. At $n = 200$, `AutoCal` achieves the highest average Spearman correlation of 0.905. At $n = 400$, `LinearCal` achieves a Spearman correlation of 0.941 and identifies the top-ranked model in 64.6% of repetitions; see Appendix K.4.

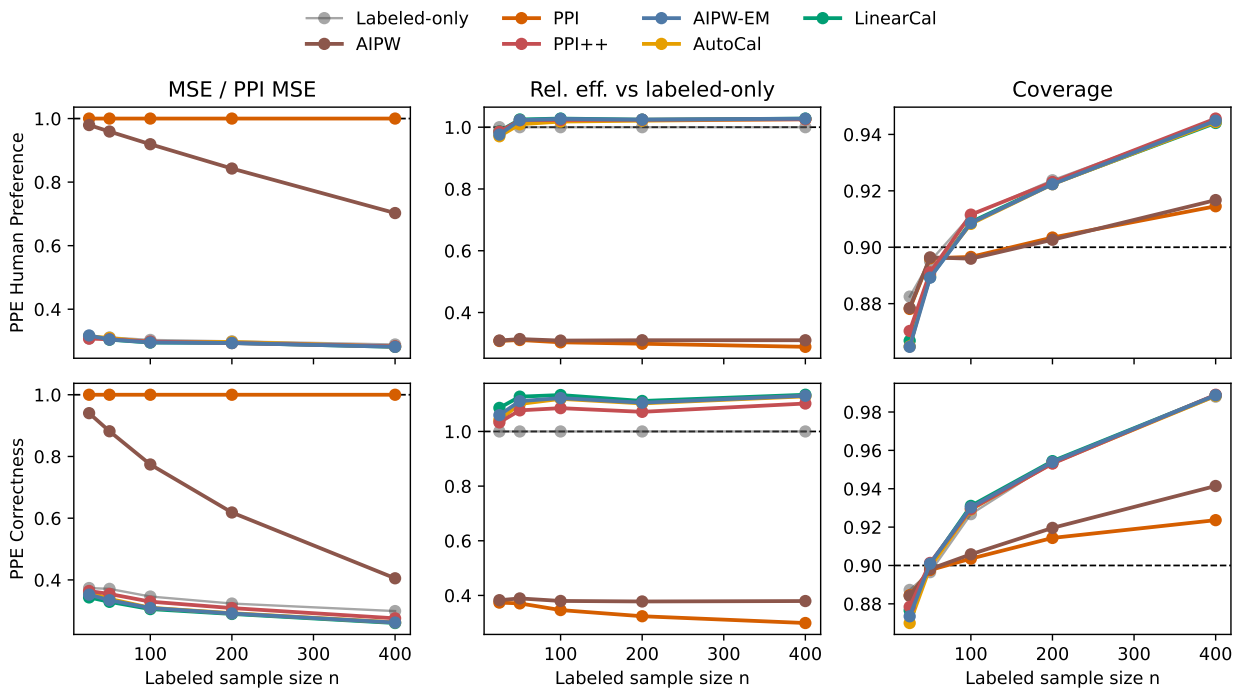


Figure 4: LLM evaluation results on PPE Human Preference and PPE Correctness. Panels show MSE normalized by that of PPI, relative efficiency compared with labeled-only estimation, and coverage of nominal 90% confidence intervals. Results are averaged across the public evaluator models.

For MATH scores from ArmoRM, empirical correctness increases across twelve score bins, from 0.078 in the lowest bin to 0.920 in the highest. Platt calibration fitted on 100 labels reduces prediction MSE from 0.4683 to 0.2020 on the held-out observations; see Appendix K.2.

6 Conclusion

We introduced *calibrated prediction-powered inference*, a framework for semisupervised mean estimation based on three steps: obtain a prediction score, calibrate it using the labeled sample, and average the calibrated predictions over the pooled covariate sample. This plug-in procedure fits naturally within the broader literature on semiparametric debiasing and can be viewed as a semisupervised specialization of calibrated debiased machine learning (van der Laan et al., 2024c).

Our results show that calibration can improve semisupervised efficiency without retraining the original model, and that isotonic calibration can *calibeat* the original score and simpler post-processing rules at first order. Moreover, replacing the fitted isotonic score by its conditional-mean oracle yields no additional first-order variance reduction. Empirically, when the original score is already well calibrated, post-processing offers little improvement, whereas under substantial miscalibration, even simple linear or monotone calibration can improve efficiency while retaining a simple plug-in form.

More broadly, our results place prediction-powered inference within the established literature on semiparametric estimation, debiased machine learning, and covariate adjustment in randomized trials and missing-at-random settings. In particular, we show that PPI++ is an AIPW estimator based on empirical efficiency maximization (Rubin and van der Laan, 2008) and, when unconstrained, is first-order equivalent to linear calibration and classical prognostic-score regression adjustment (Hansen, 2008; Lin, 2013; Schuler et al., 2022). It is therefore closely related to randomized-trial methods that learn prognostic scores from larger historical, external-control, or auxiliary datasets and use them for regression adjustment in smaller trials (Hansen, 2008; Moore and

Track	n	Estimator	MSE / PPI	Label savings	Coverage
PPE Correctness	100	Labeled-only	0.346	0.000	0.927
PPE Correctness	100	PPI	1.000	0.010	0.904
PPE Correctness	100	AIPW	0.774	0.009	0.906
PPE Correctness	100	LinearCal	0.305	0.101	0.931
PPE Correctness	100	AutoCal	0.310	0.094	0.929
PPE Correctness	400	Labeled-only	0.299	0.000	0.988
PPE Correctness	400	PPI	1.000	0.020	0.924
PPE Correctness	400	AIPW	0.405	0.009	0.941
PPE Correctness	400	LinearCal	0.260	0.105	0.989
PPE Correctness	400	AutoCal	0.263	0.102	0.988
PPE Human Preference	100	Labeled-only	0.303	0.000	0.911
PPE Human Preference	100	PPI	1.000	0.004	0.897
PPE Human Preference	100	AIPW	0.919	0.004	0.896
PPE Human Preference	100	LinearCal	0.295	0.031	0.909
PPE Human Preference	100	AutoCal	0.297	0.028	0.908
PPE Human Preference	400	Labeled-only	0.289	0.000	0.945
PPE Human Preference	400	PPI	1.000	0.005	0.915
PPE Human Preference	400	AIPW	0.703	0.004	0.917
PPE Human Preference	400	LinearCal	0.282	0.028	0.944
PPE Human Preference	400	AutoCal	0.282	0.027	0.944

Table 1: LLM evaluation results at $n \in \{100, 400\}$. Reported are MSE normalized by that of PPI, implied label savings relative to labeled-only estimation, and coverage of nominal 90% Wald intervals. Results for all estimators are reported in Appendix 3.

van der Laan, 2009a; Rosenblum and van der Laan, 2009; Benkeser et al., 2021; Højbjerre-Frandsen et al., 2025). This connection suggests that semisupervised inference could benefit from closer integration with the missing-data, causal-inference, and debiased machine-learning literatures. These areas address closely related problems of using flexible regression adjustment learned from auxiliary data while retaining valid inference for the target parameter. Further progress may come from adapting methods developed for randomized trials and missing-at-random settings to semisupervised estimation (Petersen and van der Laan, 2014; Dang et al., 2023; Ho et al., 2023; Smith et al., 2023; Ho et al., 2024).

Beyond mean estimation. These connections extend beyond the semisupervised mean-estimation setting considered here. When outcomes or covariates are missing depending on observed variables, or in settings involving right-censored time-to-event outcomes and longitudinal data, existing semiparametric methods provide a natural starting point for extending calibrated estimation (Robins et al., 1994; van der Laan and Robins, 2003; Bang and Robins, 2005; Moore and van der Laan, 2009b; Rose and van der Laan, 2011; Van der Laan and Rose, 2018). More generally, M-estimation and estimating-equation methods for missing-at-random problems are well developed (Robins et al., 1994; van der Laan and Robins, 2003; Chernozhukov et al., 2018a). When missingness depends on unobserved outcomes even after conditioning on observed data, however, naively combining labeled and unlabeled data can introduce bias. Related work addresses the combination of gold-standard and potentially biased auxiliary data using semiparametric and machine-learning methods (Kallus et al., 2018; Rosenman, 2025; Dang et al., 2025; van der Laan et al., 2026). When the unlabeled sample is selected adaptively, the problem is also related to two-phase sampling, informative sampling, and adaptive data collection (van der Laan, 2008; Chow and Chang, 2008; Rose and van der Laan, 2011; Malenica et al., 2021; Zrnic and Candès, 2024b; Zhang and van der Laan, 2025; Liu et al., 2025). Extending calibrated inference to these missing-data and adaptive-sampling settings is a natural direction for future work.

While we focus on calibration of regression functions and inference on means, the ideas here may extend to broader supervised learning problems defined through loss minimization, such as median and quantile regression (Rosenblum and van der Laan, 2009; Noarov and Roth, 2023; Jung et al., 2021; Roth, 2022; Whitehouse et al., 2024). They also extend to continuous linear functionals of a regression function or M-estimand through calibrated DML (van der Laan et al., 2024c, 2025a). More generally, when the target parameter is defined by a moment equation, as in M-estimation, one could debias or calibrate the associated augmented moment equation (van der Laan and Robins, 2003; Chernozhukov et al., 2018a); see Ji et al. (2025). A practical challenge in this setting is that the augmented loss may be nonconvex or may require separate debiasing at each parameter value, which can complicate optimization. In such cases, a more practical alternative may be a multiaccuracy-style adjustment rather than explicit augmentation or calibration, as in the Efficient Plug-in Learning framework of van der Laan et al. (2024a), which addresses this issue for nonconvex Neyman-orthogonal loss functions (Foster and Syrgkanis, 2023).

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A Semiparametric background

In the unrestricted two-sample model, setting $f = \mu_0$ in Proposition 1 gives the efficient influence pair (Robins et al., 1995; Robins and Rotnitzky, 1995; Tsiatis, 2006). Regression adjustment and TMLE can yield the same first-order representation by updating the regression so that the empirical residual mean vanishes (Scharfstein et al., 1999; Van Der Laan and Rubin, 2006; Rosenblum and van der Laan, 2009). The calibration identity (5) gives an exact instance of this construction.

B Additional remarks

Remark 3 (Empirical efficiency maximization over a prediction class)**.** The empirical variance criterion $\widehat{\sigma}_n^2(f)$ in Section 2.2 can be minimized over any candidate class $\mathcal{G} \subset L^2(P_{0,X})$, rather than only over rescalings of one score:

$$\hat{f} \in \arg\min_{f \in \mathcal{G}} \widehat{\sigma}_n^2(f), \qquad \widehat{\psi}_{\text{eem}} := \widehat{\psi}(\hat{f}).$$

For a fixed score, Proposition 1 gives the population criterion

$$V(f) = \rho_0 \text{Var}[Y] + (1-\rho_0)\text{Var}[Y - f(X)].$$

Writing $\tilde{f} = f - \mathbb{E}_{P_{0,X}} f + \psi_0$, we have $\text{Var}[Y - f(X)] = \mathbb{E}_{P_0}[(Y - \tilde{f}(X))^2]$. If $\mathcal{G}$ is closed under constant shifts, population least-squares regression therefore targets a variance-minimizing score. For tuning, average held-out estimates of the influence-function variance across folds and use the selected class or tuning parameter in a cross-fitted AIPW estimator (Rubin and van der Laan, 2008).

Remark 4 (Shrinkage via Venn–Abers)**.** When the labeled sample is small, isotonic calibration may be unstable. A natural alternative is Venn–Abers calibration, originally introduced for binary classification (Vovk and Petej, 2012); see van der Laan and Alaa (2024, 2025) for generalized versions beyond the binary setting. Assume $Y \in [0,1]$, or rescale otherwise. For each x, augment the labeled calibration sample with the hypothetical point $(m(x), y)$ for $y \in \{0,1\}$, fit the resulting isotonic calibrators $f_n^{(x,0)}$ and $f_n^{(x,1)}$, and form the interval-valued prediction

$$\left[f_n^{(x,0)}\{m(x)\},\ f_n^{(x,1)}\{m(x)\}\right].$$

For the Venn–Abers comparator, we shrink the interval midpoint toward the unadjusted AIPW estimator:

$$m^\star_{n,\text{VA}}(x) := m^\star_{n,\text{mid}}(x) + \left\{f_n^{(x,1)}\{m(x)\} - f_n^{(x,0)}\{m(x)\}\right\}\left\{\widehat{\psi}_{AIPW} - m^\star_{n,\text{mid}}(x)\right\},$$

where

$$m^\star_{n,\text{mid}}(x) := \frac{f_n^{(x,1)}\{m(x)\} + f_n^{(x,0)}\{m(x)\}}{2}.$$

The shrinkage weight is the width of the Venn–Abers interval.

Remark 5 (Intercept-only calibration)**.** Set $\hat{a}_{\text{const}} := \mathbb{P}_n^L\{Y - m(X)\}$ and $m^\star_{n,\text{const}} := m + \hat{a}_{\text{const}}$. Then PPI is the unlabeled mean $\mathbb{P}_N^U m^\star_{n,\text{const}}$, whereas AIPW is the pooled mean $\rho_n \mathbb{P}_n^L m^\star_{n,\text{const}} + (1-\rho_n)\mathbb{P}_N^U m^\star_{n,\text{const}}$.

C Calibration in randomized trials

Let $A \in \{0,1\}$ denote randomized treatment and $Y = AY(1) + (1-A)Y(0)$. Assume consistency, no interference, and $A \perp (X, Y(1), Y(0))$, with positive assignment probabilities. When estimating $\mu_a := \mathbb{E}[Y(a)]$, outcomes from arm a are labeled, and covariates from the other arm form the unlabeled sample. Let $\mathbb{P}_{n,a}$ denote the empirical mean in arm a, $\mathbb{P}_n$ the pooled mean, and $\hat{\pi}_a = n_a/n$. If the calibrated arm-specific score satisfies $\mathbb{P}_{n,a}\{Y - m^\star_{n,a}(X)\} = 0$, then

$$\hat{\mu}_a^{\text{cal}} := \mathbb{P}_n\{m^\star_{n,a}(X)\} \tag{15}$$

$$= \mathbb{P}_n\{m^\star_{n,a}(X)\} + \mathbb{P}_n\left[\frac{\mathbb{1}(A=a)}{\hat{\pi}_a}\{Y - m^\star_{n,a}(X)\}\right]. \tag{16}$$

The residual correction is zero because it equals $\mathbb{P}_{n,a}\{Y - m^\star_{n,a}(X)\}$. Thus (5) applies to each arm, and the calibrated treatment-effect estimator is

$$\hat{\tau}^{\text{cal}} := \hat{\mu}_1^{\text{cal}} - \hat{\mu}_0^{\text{cal}}. \tag{17}$$

These arm-specific updates are instances of calibrated DML (van der Laan et al., 2024c).

D AIPW comparison and balancing representations

Lemma 1 (AIPW comparison identity)**.** *Suppose $\mathbb{P}_n^L\{Y - m_n^\star(X)\} = 0$. Then, for any comparison score g for which the empirical averages are finite,*

$$\widehat{\psi}_{\text{cal}} = \widehat{\psi}(g) + (1 - \rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)(m_n^\star - g).$$

Proof. By (5), $\widehat{\psi}_{\text{cal}} = \widehat{\psi}(m_n^\star)$. Subtracting $\widehat{\psi}(g)$ and collecting terms gives the stated identity. □

Corollary 3 (Balancing-weight representation)**.** *Let $\mathcal{F}$ contain the identity map and suppose $\mathbb{P}_n^L[f(m_n^\star)\{Y - m_n^\star\}] = 0$ for every $f \in \mathcal{F}$. If $\hat{w} \in \mathcal{F}$ satisfies*

$$\mathbb{P}_n^L[\hat{w}(m_n^\star)f(m_n^\star)] = \rho_n \mathbb{P}_n^L f(m_n^\star) + (1 - \rho_n)\mathbb{P}_N^U f(m_n^\star), \qquad f \in \mathcal{F},$$

then $\widehat{\psi}_{\text{cal}} = \mathbb{P}_n^L[\hat{w}(m_n^\star)Y]$.

Proof. Taking $f(t) = t$ in the balance equation gives $\mathbb{P}_n^L[\hat{w}(m_n^\star)m_n^\star] = \widehat{\psi}_{\text{cal}}$. Residual orthogonality with $f = \hat{w}$ gives $\mathbb{P}_n^L[\hat{w}(m_n^\star)(Y - m_n^\star)] = 0$, proving the identity. □

E PPI++ and linear calibration

Proof of Proposition 3. Write

$$\hat{v}_L := \mathbb{P}_n^L[(m - \mathbb{P}_n^L m)^2], \quad \hat{v}_U := \mathbb{P}_N^U[(m - \mathbb{P}_N^U m)^2], \quad \hat{c}_L := \mathbb{P}_n^L[(Y - \mathbb{P}_n^L Y)(m - \mathbb{P}_n^L m)].$$

On the event $\hat{v}_L > 0$, the least-squares normal equations give

$$\hat{a}_{\text{lin}} = \frac{\hat{c}_L}{\hat{v}_L}, \qquad \hat{b}_{\text{lin}} = \mathbb{P}_n^L Y - \hat{a}_{\text{lin}} \mathbb{P}_n^L m.$$

Differentiating the empirical variance criterion in Section 2.2 gives

$$\hat{\lambda}_{++} = \frac{\hat{c}_L}{(1 - \rho_n)\hat{v}_L + \rho_n \hat{v}_U}.$$

Finite second moments and $\text{Var}[m(X)] > 0$ imply $\hat{v}_L, \hat{v}_U \to_p \text{Var}[m(X)]$ and $\hat{c}_L \to_p \text{Cov}(Y, m(X))$. Thus $\Pr(\hat{v}_L > 0) \to 1$, both coefficients converge to $a_0 := \text{Cov}(Y, m(X))/\text{Var}[m(X)]$, and $\hat{\lambda}_{++} - \hat{a}_{\text{lin}} = o_p(1)$.

For $\Delta_n := \mathbb{P}_N^U m - \mathbb{P}_n^L m$, substituting the fitted intercept gives

$$\widehat{\psi}_{\text{lin}} = \mathbb{P}_n^L Y + (1 - \rho_n)\hat{a}_{\text{lin}}\Delta_n, \qquad \widehat{\psi}_{++} = \mathbb{P}_n^L Y + (1 - \rho_n)\hat{\lambda}_{++}\Delta_n.$$

Subtracting proves the exact identity in the proposition. Independence and finite second moments give $\Delta_n = O_p(n^{-1/2} + N^{-1/2})$. Since $\rho_n \to \rho_0 < 1$, the ratio n/N is bounded, so

$$\widehat{\psi}_{++} - \widehat{\psi}_{\text{lin}} = (1 - \rho_n)o_p(1)\Delta_n = o_p(n^{-1/2}).$$

□

F Proof of Proposition 1

Proof of Proposition 1. Adding a constant to f leaves $\widehat{\psi}(f)$ unchanged. Thus $\widehat{\psi}(f) = \widehat{\psi}(\tilde{f})$, where $\tilde{f} = f - \mathbb{E}_{P_{0,X}} f + \psi_0$. Centering the empirical means gives the exact identity

$$\widehat{\psi}(f) - \psi_0 = (\mathbb{P}_n^L - P_0) D_{f,n}^L + (\mathbb{P}_N^U - P_{0,X}) D_{f,n}^U.$$

Both components have mean zero. Write $A = \mu_0 - \psi_0$, $\delta = \tilde{f} - \mu_0$, and $\varepsilon = Y - \mu_0$. Then

$$D_{f,n}^L = \varepsilon + \rho_n A - (1 - \rho_n)\delta, \qquad D_{f,n}^U = (1 - \rho_n)(A + \delta).$$

Since $\mathbb{E}[\varepsilon \mid X] = 0$ and $n/N = \rho_n/(1 - \rho_n)$, the covariance terms cancel in

$$\mathrm{Var}(D_{f,n}^L) + \frac{n}{N}\mathrm{Var}(D_{f,n}^U) = \mathbb{E}[\mathrm{Var}(Y \mid X)] + \rho_n \mathrm{Var}(\mu_0) + (1 - \rho_n)\mathrm{Var}(\delta).$$

Taking limits gives the stated variance, since $\mathrm{Var}(\delta) = \mathrm{Var}(f - \mu_0)$. The choice $f = \mu_0$ removes the last term and gives the missing-data efficiency bound (Robins et al., 1995; Robins and Rotnitzky, 1995; Tsiatis, 2006). □

G Supporting lemmas for isotonic calibration

Lemma 2 (Blockwise residual orthogonality)**.** *Let $B_1, \ldots, B_J$ be the pooled adjacent violator blocks of the isotonic fit, and let $m_{n,\mathrm{iso}}^\star(X_i) = c_j$ for $i \in B_j$. Then*

$$\sum_{i \in B_j} (Y_i - c_j) = 0, \qquad j = 1, \ldots, J.$$

Consequently, for every measurable function $h : \mathbb{R} \to \mathbb{R}$ such that $h(m_{n,\mathrm{iso}}^\star(X_i))$ is finite for all i,

$$\sum_{i=1}^n h(m_{n,\mathrm{iso}}^\star(X_i))\{Y_i - m_{n,\mathrm{iso}}^\star(X_i)\} = 0.$$

Proof. The fitted values from isotonic regression are constant on pooled adjacent violator blocks. The KKT conditions imply that residuals sum to zero on each block. Since $h(m_{n,\mathrm{iso}}^\star(X_i)) = h(c_j)$ for all $i \in B_j$,

$$\sum_{i=1}^n h(m_{n,\mathrm{iso}}^\star(X_i))\{Y_i - m_{n,\mathrm{iso}}^\star(X_i)\} = \sum_{j=1}^J h(c_j) \sum_{i \in B_j} (Y_i - c_j) = 0.$$

□

Proof of proposition 2. Lemma 2 gives (6). By definition, the fitted isotonic score minimizes the empirical squared loss among all nondecreasing transformations of m, proving the risk comparison. Taking the identity transformation gives the comparison with the raw score. □

Lemma 3 (Bounded variation after isotonic post-processing)**.** *Suppose Assumption 3 holds. Let $T := m(X)$ and $S_n := m_{n,\mathrm{iso}}^\star(X)$. Define*

$$\eta(t) := \mathbb{E}_{P_0}[Y \mid T = t], \qquad \eta_n(s) := \mathbb{E}_{P_0}[Y \mid S_n = s].$$

If $t \mapsto \eta(t)$ has total variation at most V, then, conditional on the labeled sample, the map $s \mapsto \eta_n(s)$ has total variation at most V. If, in addition, $|\eta(t)| \le C_0$ on the support of T, then $|\eta_n(s)| \le C_0$ on the support of S_n.

Proof. We adapt the proof of Lemma 6 in van der Laan et al. (2023). Let $S_n = f_n(T)$, where f_n is the fitted isotonic map. Conditional on the labeled sample, f_n is nondecreasing. Hence, for each s in the support of S_n, there exists an interval $B_s \subset \mathbb{R}$, possibly a singleton, such that

$$\{S_n = s\} = \{T \in B_s\}.$$

Therefore

$$\eta_n(s) = \mathbb{E}_{P_0}[Y \mid S_n = s] = \mathbb{E}_{P_0}[\eta(T) \mid T \in B_s].$$

Let $\eta = \eta^+ - \eta^-$ be the Jordan decomposition of η, where η^+ and η^- are nondecreasing and

$$\|\eta\|_{TV} = \|\eta^+\|_{TV} + \|\eta^-\|_{TV} \le V.$$

Then

$$\eta_n(s) = \mathbb{E}_{P_0}[\eta^+(T) \mid T \in B_s] - \mathbb{E}_{P_0}[\eta^-(T) \mid T \in B_s].$$

If $s_1 < s_2$, then monotonicity of f_n implies every point of B_{s_1} is no larger than every point of B_{s_2}. Hence the conditional distribution of $T \mid T \in B_{s_2}$ first-order stochastically dominates that of $T \mid T \in B_{s_1}$. Therefore, for any nondecreasing function g,

$$\mathbb{E}_{P_0}[g(T) \mid T \in B_{s_1}] \le \mathbb{E}_{P_0}[g(T) \mid T \in B_{s_2}].$$

Applying this with $g = \eta^+$ and $g = \eta^-$, both maps

$$s \mapsto \mathbb{E}_{P_0}[\eta^+(T) \mid T \in B_s] \qquad \text{and} \qquad s \mapsto \mathbb{E}_{P_0}[\eta^-(T) \mid T \in B_s]$$

are nondecreasing.

Moreover, each of these conditional expectations takes values in the range of the corresponding function, so

$$\big\|s \mapsto \mathbb{E}_{P_0}[\eta^+(T) \mid T \in B_s]\big\|_{TV} \le \|\eta^+\|_{TV}, \qquad \big\|s \mapsto \mathbb{E}_{P_0}[\eta^-(T) \mid T \in B_s]\big\|_{TV} \le \|\eta^-\|_{TV}.$$

Hence

$$\|\eta_n\|_{TV} \le \|\eta^+\|_{TV} + \|\eta^-\|_{TV} = \|\eta\|_{TV} \le V.$$

Finally, if $|\eta(t)| \le C_0$ on the support of T, then

$$|\eta_n(s)| = |\mathbb{E}_{P_0}[\eta(T) \mid T \in B_s]| \le C_0$$

for all s in the support of S_n. □

Lemma 4 (The population isotonic projection is centered)**.** *Under Assumptions 1 and 2,* $\mathbb{E}_{P_{0,X}}[m_0(X)] = \psi_0$.

Proof. The isotonic class is closed under constant shifts, so $c \mapsto \mathbb{E}[(Y - m_0(X) - c)^2]$ is minimized at $c = 0$. Its derivative there is $-2\mathbb{E}[Y - m_0(X)]$, giving the stated identity. □

Proof convention. In the following proofs, fix $\zeta > 0$. By $\sup_x |m^\star_{n,\mathrm{iso}}(x)| = O_p(1)$, there exist a deterministic constant $C = C_\zeta \ge C_0$ and $n_0 < \infty$ such that the event

$$A_{n,C} := \Big\{\sup_x |m^\star_{n,\mathrm{iso}}(x)| \le C\Big\}$$

satisfies $P_0(A_{n,C}) \ge 1 - \zeta$ for all $n \ge n_0$. We carry out the deterministic bounded-class arguments on $A_{n,C}$, where all implicit constants may depend on C but not on n. Since ζ is arbitrary, the resulting rates and distributional statements hold unconditionally.

Lemma 5 (Isotonic L^2 rate)**.** *Suppose Assumptions 1 and 2 holds. Then*

$$\|m^\star_{n,\mathrm{iso}} - m_0\|^2_{2,P_{0,X}} = O_p(n^{-2/3}).$$

Proof of Lemma 5. Let $T := m(X)$, $\eta(T) := \mathbb{E}_{P_0}[Y \mid T]$, and

$$\mathcal{M}_C := \{f(T) : f \in \mathcal{F}_{\text{iso}},\ \|f(T)\|_\infty \le C\}.$$

By Assumption 2, $\eta(T)$ is bounded by $C_0 \le C$, and on $A_{n,C}$ we have $m^\star_{n,\text{iso}} \in \mathcal{M}_C$. Since clipping any monotone candidate to $[-C, C]$ preserves monotonicity and can only decrease squared risk against $\eta(T)$, the population isotonic projection m_0 also belongs to $\mathcal{M}_C$. For every $m \in \mathcal{M}_C$,

$$\mathbb{E}_{P_0}[(Y - m)^2] = \mathbb{E}_{P_0}[(Y - \eta)^2] + \mathbb{E}_{P_0}[(\eta - m)^2],$$

which shows that m_0 is the $L^2(P_T)$-projection of η onto the closed convex set $\mathcal{M}_C$.

Now define the centered class

$$\mathcal{F}_0 := \{g = m - m_0 :\ m \in \mathcal{M}_C\}.$$

For $g \in \mathcal{F}_0$, let $\ell_g(Z) := (Y - m_0(X) - g(X))^2 - (Y - m_0(X))^2$, where $Z = (X, Y)$. Since $m^\star_{n,\text{iso}}$ minimizes the empirical squared risk over $\mathcal{F}_{\text{iso}}$, on $A_{n,C}$ the random element $\hat{g} := m^\star_{n,\text{iso}} - m_0$ belongs to $\mathcal{F}_0$ and satisfies

$$\mathbb{P}_n^L\{\ell_{\hat{g}}\} \le 0 = \mathbb{P}_n^L\{\ell_0\}.$$

Moreover,

$$\mathbb{E}_{P_0}[\ell_g] = \|g\|^2_{2,P_{0,X}} - 2\mathbb{E}_{P_0}[(Y - m_0)g].$$

Because g is measurable with respect to T, the tower property gives

$$\mathbb{E}_{P_0}[(Y - m_0)g] = \mathbb{E}_{P_0}[(\eta - m_0)g].$$

Since m_0 is the $L^2(P_T)$-projection of η onto the convex set $\mathcal{M}_C$, the Hilbert-space projection inequality yields

$$\mathbb{E}_{P_0}[(\eta - m_0)(m - m_0)] \le 0 \qquad \text{for every } m \in \mathcal{M}_C.$$

Hence

$$\mathbb{E}_{P_0}[\ell_g] \ge \|g\|^2_{2,P_{0,X}},$$

so squared loss has the required quadratic margin around m_0.

We next bound the modulus of continuity of the empirical process indexed by the localized classes

$$\mathcal{F}_\delta := \{g \in \mathcal{F}_0 : \|g\|_{2,P_{0,X}} \le \delta\}.$$

The bracketing bound for bounded monotone functions gives

$$\log N_{[]}(\varepsilon, \mathcal{F}_\delta, L_2(P_{0,X})) \lesssim \varepsilon^{-1}, \qquad 0 < \varepsilon < \delta.$$

Consequently,

$$J_{[]}(\delta, \mathcal{F}_\delta, L_2(P_{0,X})) \lesssim \delta^{1/2};$$

see, e.g., van der Vaart and Wellner (1996).

Write $\xi := Y - m_0(X)$, so that $\ell_g = g^2 - 2\xi g$. By Condition (ii) and the bounds on η, m_0, and g, there exists $A_C < \infty$ such that

$$\mathbb{E}_{P_0}[|\xi g|^k] \le k! A_C^k \|g\|^2_{2,P_{0,X}}, \qquad g \in \mathcal{F}_0, \quad k \ge 2. \tag{18}$$

The same inequality holds for bounded bracket widths. Thus multiplication by ξ maps $L^2(P_{0,X})$ brackets to Bernstein-norm brackets with widths bounded by a constant multiple of their original widths; see Lemmas 8–9 of Bibaut and van der Laan (2019). The squared class $\{g^2 : g \in \mathcal{F}_\delta\}$ is a bounded Lipschitz image of $\mathcal{F}_\delta$. Applying Lemmas 3.4.2–3.4.3 of van der Vaart and Wellner (1996) to these two classes gives

$$E^\star \sup_{g \in \mathcal{F}_\delta} |(\mathbb{P}_n^L - P_0)\ell_g| \lesssim n^{-1/2}\delta^{1/2} + n^{-1}\delta^{-1}. \tag{19}$$

For $\delta \ge n^{-1/3}$, the right-hand side is bounded by a constant times $n^{-1/2}\delta^{1/2}$.

Set $\varepsilon_n := n^{-1/3}$. The empirical risk inequality, the quadratic margin, and (19) imply, by peeling into shells $2^s\varepsilon_n \le \|\hat{g}\|_{2,P_{0,X}} < 2^{s+1}\varepsilon_n$ and applying Markov's inequality, that

$$\begin{aligned}\Pr\{\|\hat{g}\|_{2,P_{0,X}} \ge 2^S\varepsilon_n,\ A_{n,C}\} &\lesssim \sum_{s\ge S} \frac{n^{-1/2}(2^{s+1}\varepsilon_n)^{1/2}}{2^{2s}\varepsilon_n^2} \\ &\lesssim \sum_{s\ge S} 2^{-3s/2} \longrightarrow 0 \qquad (S\to\infty).\end{aligned}$$

Hence

$$\|m^\star_{n,\mathrm{iso}} - m_0\|_{2,P_{0,X}} = O_p(n^{-1/3}), \qquad \|m^\star_{n,\mathrm{iso}} - m_0\|^2_{2,P_{0,X}} = O_p(n^{-2/3}).$$

Since $P_0(A^c_{n,C}) \le \zeta$ for all large n, and $\zeta > 0$ was arbitrary, the same rate holds unconditionally, proving Lemma 5. □

Theorem 3 (Calibration error of isotonic post-processing)**.** *Suppose Assumptions 1 to 3 hold. Let*

$$m_{0|n}(X) := \mathbb{E}_{P_0}[Y \mid m^\star_{n,\mathrm{iso}}(X)].$$

Then

$$\|m_{0|n} - m^\star_{n,\mathrm{iso}}\|^2_{2,P_{0,X}} = O_p(n^{-2/3}).$$

Proof. The proof follows that of Theorem 1 in van der Laan et al. (2023); see also van der Laan and Alaa (2025). Let

$$T := m(X), \qquad S_n := m^\star_{n,\mathrm{iso}}(X), \qquad \eta_n(s) := \mathbb{E}_{P_0}[Y \mid S_n = s],$$

and define

$$\Delta_n(X) := m_{0|n}(X) - m^\star_{n,\mathrm{iso}}(X) = \eta_n(S_n) - S_n.$$

Conditional on the labeled sample, $h_n(s) := \eta_n(s) - s$ is a measurable function of s. Applying proposition 2 pathwise with $h = h_n$ yields

$$\mathbb{P}^L_n\{\Delta_n(X)\{Y - S_n\}\} = 0.$$

Also,

$$\mathbb{E}_{P_0}[\Delta_n(X)\{Y - S_n\}] = \mathbb{E}_{P_0}[\Delta_n(X)\,\mathbb{E}_{P_0}[Y - S_n \mid S_n]] = \mathbb{E}_{P_0}[\Delta_n^2] = \|\Delta_n\|^2_{2,P_{0,X}}.$$

Hence

$$\|\Delta_n\|^2_{2,P_{0,X}} = (\mathbb{E}_{P_0} - \mathbb{P}^L_n)\{\Delta_n(X)\{Y - S_n\}\}.$$

Fix $\zeta > 0$, and work on the event $A_{n,C}$ from the proof convention. On this event, S_n is uniformly bounded by C. By Lemma 3, the function η_n belongs to the bounded-variation class

$$\mathcal{F}_{\mathrm{TV},C} := \{f : \mathbb{R} \to \mathbb{R} : \|f\|_\infty \le C,\ \|f\|_{TV} \le V\},$$

where $V < \infty$ is the variation bound in Assumption 3. Let

$$\mathcal{F}_{\mathrm{iso},C} := \{f : \mathbb{R} \to \mathbb{R} : f \text{ nondecreasing and } \|f\|_\infty \le C\}.$$

Since $S_n = f_n(T)$ for a nondecreasing fitted isotonic map f_n, and composition with a monotone map preserves bounded variation, the function

$$t \mapsto \eta_n\{f_n(t)\}$$

also belongs to $\mathcal{F}_{\mathrm{TV},C}$, after enlarging C if necessary. Thus $\eta_n(S_n) = h_1(X)$ for some $h_1 = f_1 \circ m$ with $f_1 \in \mathcal{F}_{\mathrm{TV},C}$, while $S_n = h_2(X)$ for $h_2 = f_2 \circ m$ with $f_2 = f_n \in \mathcal{F}_{\mathrm{iso},C}$.

Define the localized product class

$$\mathcal{G}_{n,C,\delta} := \Big\{(h_1 - h_2)(Y - h_2) : h_1 = f_1 \circ m,\ h_2 = f_2 \circ m,\ f_1 \in \mathcal{F}_{\mathrm{TV},C},\ f_2 \in \mathcal{F}_{\mathrm{iso},C},\ \|h_1 - h_2\|_{2,P_{0,X}} \le \delta\Big\}.$$

Then

$$\Delta_n(X)\{Y - S_n\} \in \mathcal{G}_{n,C,\delta_n}, \qquad \delta_n := \|\Delta_n\|_{2,P_{0,X}}.$$

Write $\eta(T) := \mathbb{E}_{P_0}[Y \mid T]$ and $\varepsilon := Y - \eta(T)$. Each element of $\mathcal{G}_{n,C,\delta}$ decomposes as

$$(h_1 - h_2)(Y - h_2) = (h_1 - h_2)\varepsilon + (h_1 - h_2)\{\eta(T) - h_2\}.$$

Bounded-variation functions are differences of bounded monotone functions, so their bracketing entropy is $O(\epsilon^{-1})$. The second term is a bounded Lipschitz transformation of (h_1, h_2). For the first term, Condition (ii) gives the moment bound in (18), with $h_1 - h_2$ in place of g and ε in place of ξ. Its Bernstein-norm bracket widths are therefore controlled by $L^2(P_{0,X})$ bracket widths. The second term has $L^2(P_0)$ norm $O(\delta)$, and the first has Bernstein norm $O(\delta)$; both bracketing entropy integrals are $O(\delta^{1/2})$. Lemmas 3.4.2–3.4.3 of van der Vaart and Wellner (1996) give

$$E^\star \sup_{g \in \mathcal{G}_{n,C,\delta}} |(\mathbb{P}_n^L - P_0)\{g\}| \lesssim n^{-1/2}\delta^{1/2} + n^{-1}\delta^{-1}.$$

For $\delta \geq n^{-1/3}$, this is bounded by a constant times $n^{-1/2}\delta^{1/2}$.

Set $\varepsilon_n := n^{-1/3}$. For $s \geq 0$, let

$$B_s := \{2^s\varepsilon_n \leq \delta_n < 2^{s+1}\varepsilon_n\}.$$

On $B_s \cap A_{n,C}$,

$$2^{2s}\varepsilon_n^2 \leq \delta_n^2 \leq \sup_{g \in \mathcal{G}_{n,C,2^{s+1}\varepsilon_n}} |(\mathbb{P}_n^L - P_0)\{g\}|.$$

Therefore, by Markov's inequality,

$$P_0(B_s \cap A_{n,C}) \lesssim \frac{E^\star \sup_{g \in \mathcal{G}_{n,C,2^{s+1}\varepsilon_n}} |(\mathbb{P}_n^L - P_0)\{g\}|}{2^{2s}\varepsilon_n^2} \lesssim \frac{n^{-1/2}(2^{s+1}\varepsilon_n)^{1/2}}{2^{2s}\varepsilon_n^2} \lesssim 2^{-3s/2},$$

since $n^{-1/2}\varepsilon_n^{-3/2} = 1$. Summing over $s \geq S$ yields

$$P_0(\delta_n \geq 2^S\varepsilon_n,\ A_{n,C}) \lesssim \sum_{s=S}^{\infty} 2^{-3s/2} \xrightarrow[S\to\infty]{} 0.$$

Because $P_0(A_{n,C}^c) \leq \zeta$ for all large n, and $\zeta > 0$ is arbitrary, it follows that

$$\|\Delta_n\|_{2,P_{0,X}} = O_p(n^{-1/3}), \qquad \|\Delta_n\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

This is exactly the claimed calibration-error rate. □

Lemma 6 (Centered empirical-process bounds for isotonic calibration)**.** *Suppose Assumptions 1 and 2 holds. Then*

$$(\mathbb{P}_n^L - P_{0,X})\{m^\star_{n,\mathrm{iso}}(X) - m_0(X)\} = O_p(n^{-2/3}), \qquad (\mathbb{P}_N^U - P_{0,X})\{m^\star_{n,\mathrm{iso}}(\widetilde{X}) - m_0(\widetilde{X})\} = O_p\big(n^{-1/3}N^{-1/2}\big).$$

Proof of Lemma 6. By Lemma 5,

$$\|m^\star_{n,\mathrm{iso}} - m_0\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

To control the centered empirical-process term, work on the event $A_{n,C}$ from the proof convention. For fixed $R < \infty$, consider the localized class

$$\mathcal{F}_{n,C} := \{g = f \circ m - m_0 :\ f \in \mathcal{F}_{\mathrm{iso}},\ \|f \circ m\|_\infty \leq C,\ \|g\|_{2,P_{0,X}} \leq Rn^{-1/3}\}.$$

By Lemma 5, the probability of $A_{n,C} \cap \{m^\star_{n,\mathrm{iso}} - m_0 \notin \mathcal{F}_{n,C}\}$ can be made arbitrarily small by taking R large. Because m is fixed, $\mathcal{F}_{n,C}$ is a translated class of bounded monotone functions of a one-dimensional score. The monotone bracketing bound implies

$$J_{[]}(r, \mathcal{F}_{n,C}, L_2(P_0)) \lesssim r^{1/2}.$$

Applying Lemma 3.4.2 of van der Vaart and Wellner (1996) at radius $r_n = Rn^{-1/3}$ yields

$$\sup_{g\in\mathcal{F}_{n,C}} |(\mathbb{P}_n^L - P_{0,X})\{g\}| = O_p\Big(n^{-1/2}r_n^{1/2} + n^{-1}r_n^{-1}\Big) = O_p(n^{-2/3}).$$

In particular,

$$(\mathbb{P}_n^L - P_{0,X})\{m^\star_{n,\mathrm{iso}} - m_0\} = O_p(n^{-2/3}).$$

Since $P_0(A_{n,C}^c) \le \zeta$ for all large n, and $\zeta > 0$ is arbitrary, the same bound holds unconditionally.

For the unlabeled sample, conditional on the labeled data, $m^\star_{n,\mathrm{iso}} - m_0$ is fixed and the unlabeled sample is independent, so

$$(\mathbb{P}_N^U - P_{0,X})\{m^\star_{n,\mathrm{iso}} - m_0\} = O_p\left(N^{-1/2}\|m^\star_{n,\mathrm{iso}} - m_0\|_{2,P_{0,X}}\right) = O_p\big(n^{-1/3}N^{-1/2}\big)$$

by Lemma 5. □

Lemma 7 (Centered empirical-process bounds for isotonic post-processing)**.** *Suppose Assumptions 1 to 3 hold. Let*

$$\Delta_n(X) := m_{0|n}(X) - m^\star_{n,\mathrm{iso}}(X).$$

Then

$$\big(\mathbb{P}_n^L - P_{0,X}\big)\{\Delta_n(X)\} = O_p(n^{-2/3}), \qquad \big(\mathbb{P}_N^U - P_{0,X}\big)\{\Delta_n(\widetilde{X})\} = O_p\big(n^{-1/3}N^{-1/2}\big).$$

Proof. By theorem 3,

$$\|\Delta_n\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

Let $V < \infty$ be the variation bound in Assumption 3. Fix $\zeta > 0$, and work on the event $A_{n,C}$. On this event, $m^\star_{n,\mathrm{iso}}$ is uniformly bounded by C. By Lemma 3, the function

$$s \mapsto \eta_n(s) := \mathbb{E}_{P_0}[Y \mid m^\star_{n,\mathrm{iso}}(X) = s]$$

has total variation at most V and is bounded by C on the support of $m^\star_{n,\mathrm{iso}}(X)$. Since $m^\star_{n,\mathrm{iso}}(X) = f_n(T)$ with $T = m(X)$ and f_n nondecreasing, it follows that

$$m_{0|n}(X) = \eta_n\{f_n(T)\}$$

is a bounded-variation transform of T, while $m^\star_{n,\mathrm{iso}}(X)$ is a bounded monotone transform of T.

Consider the classes

$$\mathcal{F}_{\mathrm{TV},C} := \{f : \mathbb{R} \to \mathbb{R} : \|f\|_\infty \le C,\ \|f\|_{TV} \le V\},$$
$$\mathcal{F}_{\mathrm{iso},C} := \{f : \mathbb{R} \to \mathbb{R} : f \text{ nondecreasing and } \|f\|_\infty \le C\}.$$

For fixed $R < \infty$, define

$$\mathcal{H}_{n,C} := \Big\{h_1 - h_2 : h_1 = f_1 \circ m,\ h_2 = f_2 \circ m,\ f_1 \in \mathcal{F}_{\mathrm{TV},C},\ f_2 \in \mathcal{F}_{\mathrm{iso},C},\ \|h_1 - h_2\|_{2,P_{0,X}} \le Rn^{-1/3}\Big\}.$$

By theorem 3, the probability of $A_{n,C} \cap \{\Delta_n \notin \mathcal{H}_{n,C}\}$ can be made arbitrarily small by taking R large.

Because bounded-variation functions are differences of bounded monotone functions, the localized bracketing entropy of $\mathcal{H}_{n,C}$ obeys the same bound as in the proof of theorem 3, namely

$$J_{[]}(\delta, \mathcal{H}_{n,C}, L_2(P_0)) \lesssim \delta^{1/2}.$$

Applying Lemma 3.4.2 of van der Vaart and Wellner (1996) at radius $r_n = Rn^{-1/3}$ gives

$$\sup_{h\in\mathcal{H}_{n,C}} |(\mathbb{P}_n^L - P_{0,X})h| = O_p\Big(n^{-1/2}r_n^{1/2} + n^{-1}r_n^{-1}\Big) = O_p(n^{-2/3}).$$

Taking R large gives the same bound for $(\mathbb{P}_n^L - P_{0,X})\Delta_n$ on $A_{n,C}$. Since $P_0(A_{n,C}^c) \le \zeta$ for all large n, and $\zeta > 0$ was arbitrary, the same bound holds unconditionally.

For the unlabeled sample, conditional on the labeled data, Δ_n is fixed and the unlabeled sample is independent. Hence

$$\big(\mathbb{P}_N^U - P_{0,X}\big)\{\Delta_n(\widetilde{X})\} = O_p\left(N^{-1/2}\|\Delta_n\|_{2,P_{0,X}}\right) = O_p\big(n^{-1/3}N^{-1/2}\big)$$

by theorem 3. □

H Proof of asymptotic linearity

Proof of theorem 1. By Lemma 4, $\mathbb{E}_{P_{0,X}}[m_0] = \psi_0$. The isotonic fit has mean-zero labeled residuals by Lemma 2. Applying Lemma 1 with $g = m_0$ gives

$$\widehat{\psi}_{\text{iso}} = \rho_n \mathbb{P}_n^L \left\{ m_0 + \rho_n^{-1}(Y - m_0) \right\} + (1-\rho_n)\mathbb{P}_N^U\{m_0(\widetilde{X})\} + (1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\{m^\star_{n,\text{iso}} - m_0\}.$$

Subtracting ψ_0 gives

$$\begin{aligned}\widehat{\psi}_{\text{iso}} - \psi_0 = (\mathbb{P}_n^L - P_0)D^L_{m_0,n} + (\mathbb{P}_N^U - P_{0,X})D^U_{m_0,n} \\ + (1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\{m^\star_{n,\text{iso}} - m_0\}.\end{aligned}$$

Also,

$$(1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\{m^\star_{n,\text{iso}} - m_0\} = (1-\rho_n)(\mathbb{P}_N^U - P_{0,X})\{m^\star_{n,\text{iso}} - m_0\} - (1-\rho_n)(\mathbb{P}_n^L - P_{0,X})\{m^\star_{n,\text{iso}} - m_0\}.$$

By Lemma 6, the labeled term is $O_p(n^{-2/3})$ and the unlabeled term is $O_p(n^{-1/3}N^{-1/2})$. Hence the remainder satisfies the bound in equation (14). Moreover,

$$\sqrt{n}(1-\rho_n)n^{-2/3} \le n^{-1/6}$$

and, writing $t_n = N/n$,

$$\sqrt{n}(1-\rho_n)n^{-1/3}N^{-1/2} = n^{-1/3}\frac{\sqrt{t_n}}{1+t_n} \le \frac{1}{2}n^{-1/3}.$$

Therefore $R_{n,N} = o_p(n^{-1/2})$, which proves equation (14). □

Proof of corollary 1. By theorem 1,

$$\sqrt{n}\,(\widehat{\psi}_{\text{iso}} - \psi_0) = \frac{1}{\sqrt{n}}\sum_{i=1}^n D^L_{m_0,n}(X_i, Y_i) + \frac{\sqrt{n}}{N}\sum_{j=1}^N D^U_{m_0,n}(\widetilde{X}_j) + o_p(1).$$

The labeled and unlabeled samples are independent and each summand has mean zero. Moreover,

$$\frac{n}{N}\text{Var}\{D^U_{m_0,n}(\widetilde{X})\} = \rho_n(1-\rho_n)\text{Var}\{m_0(X)\}.$$

Since $\rho_n \to \rho_0 \in [0,1)$, a triangular-array central limit theorem yields the stated asymptotic normal law with variance σ_0^2.

For the variance estimator, let $g_n := m^\star_{n,\text{iso}} - m_0$. By Lemma 5,

$$\mathbb{E}_{P_{0,X}}[g_n^2] = \|g_n\|^2_{2,P_{0,X}} = O_p(n^{-2/3}).$$

Applying Lemma 3.4.2 of van der Vaart and Wellner (1996) to the squared class gives

$$\sup_{g \in \mathcal{F}_\delta} |(\mathbb{P}_n^L - P_{0,X})\{g^2\}| = O_p\Big(n^{-1/2}\delta^{1/2} + n^{-1}\delta^{-1}\Big),$$

where $\mathcal{F}_\delta$ is the localized class from the proof of Lemma 5. By that lemma, on $A_{n,C}$ the function g_n belongs to $\mathcal{F}_{Rn^{-1/3}}$ with arbitrarily high probability for sufficiently large fixed R. Taking $\delta = Rn^{-1/3}$ gives

$$(\mathbb{P}_n^L - P_{0,X})\{g_n^2\} = O_p(n^{-2/3}).$$

Hence

$$\mathbb{P}_n^L(g_n^2) = O_p(n^{-2/3}) = o_p(1).$$

On the boundedness event $A_{n,C}$, conditional on the labeled data, g_n is fixed and $|g_n| \le 2C$, so

$$(\mathbb{P}_N^U - P_{0,X})\{g_n^2\} = O_p\left(N^{-1/2}\|g_n^2\|_{2,P_{0,X}}\right) = O_p\left(C\,N^{-1/2}\|g_n\|_{2,P_{0,X}}\right) = O_p(n^{-1/3}N^{-1/2}).$$

Therefore

$$\mathbb{P}_N^U(g_n^2) = O_p(n^{-2/3}) + O_p\big(n^{-1/3}N^{-1/2}\big) = o_p(1).$$

Since $P_0(A_{n,C}^c) \le \zeta$ for all large n and arbitrary $\zeta > 0$, these bounds hold unconditionally. Also, $\widehat{\psi}_{\text{iso}} \to_p \psi_0$ follows from theorem 1.

Finally,

$$\widehat{D}^L - D^L_{m_0,n} = -(1-\rho_n)g_n - \rho_n(\widehat{\psi}_{\text{iso}} - \psi_0),$$

and

$$\widehat{D}^U - D^U_{m_0,n} = (1-\rho_n)\{g_n(\widetilde{X}) - (\widehat{\psi}_{\text{iso}} - \psi_0)\}.$$

Because $\mathbb{P}_n^L(g_n^2), \mathbb{P}_N^U(g_n^2) = o_p(1)$ and $\widehat{\psi}_{\text{iso}} \to_p \psi_0$, it follows that

$$\mathbb{P}_n^L\Big[\{\widehat{D}^L - D^L_{m_0,n}\}^2\Big] = o_p(1), \qquad \frac{n}{N}\mathbb{P}_N^U\Big[\{\widehat{D}^U - D^U_{m_0,n}\}^2\Big] = o_p(1),$$

where the second conclusion uses

$$\frac{n}{N}(1-\rho_n)^2 = \rho_n(1-\rho_n) \le \frac{1}{4}.$$

The law of large numbers and $\rho_n \to \rho_0$ also give

$$\mathbb{P}_n^L\{(D^L_{m_0,n})^2\} + \frac{n}{N}\mathbb{P}_N^U\{(D^U_{m_0,n})^2\} \;\to_p\; \sigma_0^2.$$

The empirical means of both estimated influence components converge to zero, so within-sample centering does not change this limit. Therefore $\widehat{\sigma}^2 \to_p \sigma_0^2$. The Wald interval claim then follows by Slutsky's theorem. □

I Proof of post-processing equivalence and reduced-model efficiency

Proof of Theorem 2. Set $\Delta_n = m_{0|n} - m^\star_{n,\text{iso}}$. Lemma 1 with $g = m_{0|n}$ gives

$$\widehat{\psi}_{\text{eff}} - \widehat{\psi}_{\text{iso}} = (1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\Delta_n.$$

By Lemma 7, the right-hand side is

$$O_p((1-\rho_n)n^{-2/3}) + O_p((1-\rho_n)n^{-1/3}N^{-1/2}) = o_p(n^{-1/2}),$$

using the same bounds as in the proof of Theorem 1. The calibration-error rate is Theorem 3. Together these prove both conclusions. □

Proof of corollary 2. For part (i), if $m_0 = \mu_0$ almost surely, then the asymptotic influence pair in theorem 1 is

$$D^L_{\mu_0,n}(X,Y) = Y - \psi_0 - (1-\rho_n)\{\mu_0(X) - \psi_0\}, \qquad D^U_{\mu_0,n}(\widetilde{X}) = (1-\rho_n)\{\mu_0(\widetilde{X}) - \psi_0\}.$$

By proposition 1, this is the efficient influence pair in the unrestricted two-sample model. Therefore $\widehat{\psi}_{\text{iso}}$ is efficient for ψ_0 in the full two-sample experiment.

For part (ii), fix a square-integrable nondecreasing transformation $g = \theta \circ m$, and let $g^\circ = g - \mathbb{E}_{P_{0,X}} g + \psi_0$. Adding a constant preserves monotonicity, so population isotonic optimality and Lemma 4 give

$$\text{Var}(Y - m_0) = \mathbb{E}[(Y-m_0)^2] \le \mathbb{E}[(Y-g^\circ)^2] = \text{Var}(Y-g).$$

Proposition 1 now gives the efficiency comparison.

For part (iii), let

$$S := m(X), \qquad \eta_0(s) := \mathbb{E}_{P_0}[Y \mid S = s].$$

proposition 1 applied to the reduced two-sample experiment with scalar covariate S shows that the efficient triangular influence pair in that reduced experiment is

$$D^L_{\eta_0,n}(S,Y) = Y - \psi_0 - (1-\rho_n)\{\eta_0(S) - \psi_0\}, \qquad D^U_{\eta_0,n}(\widetilde{S}) = (1-\rho_n)\{\eta_0(\widetilde{S}) - \psi_0\}.$$

If η_0 is monotone increasing, then η_0 belongs to the isotonic class and

$$\mathbb{E}_{P_0}\left[\{Y - g(S)\}^2\right] = \mathbb{E}_{P_0}\left[\{Y - \eta_0(S)\}^2\right] + \mathbb{E}_{P_0}\left[\{\eta_0(S) - g(S)\}^2\right]$$

for every measurable g. Hence the population isotonic regression of Y on the score $S = m(X)$ is exactly $\eta_0(S)$, so

$$m_0(X) = \eta_0\{m(X)\} \qquad \text{almost surely.}$$

Therefore, the influence pair in theorem 1 coincides with the reduced-model efficient influence pair above, so $\widehat{\psi}_{\text{iso}}$ is efficient for ψ_0 in the reduced experiment indexed by $m(X)$. The isotonic class is closed under constant shifts and positive rescaling. The corresponding population first-order conditions give $\mathbb{E}[Y - m_0] = \mathbb{E}[m_0(Y - m_0)] = 0$, so $\mathrm{Cov}(Y, m_0) = \mathrm{Var}(m_0)$. Substitution into σ_0^2 gives the stated variance reduction relative to the labeled mean. □

J Pooled i.i.d. Missing-Data Formulation

When $\rho_0 > 0$, the joint two-sample setup can be embedded into a pooled i.i.d. model by introducing a sample-membership indicator $R \in \{0,1\}$, where $R = 1$ denotes labeled observations and $R = 0$ denotes unlabeled observations. In that formulation one observes i.i.d. data $O = (X, R, RY)$ with $\mathbb{P}(R = 1) = \rho_0$, and the calibrated estimator becomes

$$\widehat{\psi}_{\text{cal}} = \mathbb{P}_M(m_n^\star) + \mathbb{P}_M\left[\frac{R}{\rho_0}\{Y - m_n^\star\}\right]$$

up to the same calibration identity as in the main text. The efficient influence function in the reduced model observing only the score is

$$\mathbb{E}[Y \mid m(X)] - \psi_0 + \frac{R}{\rho_0}\{Y - \mathbb{E}[Y \mid m(X)]\},$$

which is the single-sample analogue of the efficient influence pair obtained by applying proposition 1 to the reduced model with scalar covariate $m(X)$. Thus the usual i.i.d. missing-data formulation is a convenient special case, but the main text keeps the two-dataset structure explicit.

K Additional empirical details

K.1 Small-sample comparisons and RePPI

The small-sample experiment uses $S \sim N(0,1)$, $p(S) = \{1 + \exp(-5S)\}^{-1}$, and $Y \mid S \sim \text{Bernoulli}\{p(S)\}$. Let $C(t) = \min\{0.99, \max(0.01, t)\}$, $\sigma(t) = (1 + e^{-t})^{-1}$, and $Z \sim N(0,1)$ independently of (S,Y). The mildly miscalibrated score is $C[\sigma\{1.5\,\mathrm{logit}(C[p(S) + 0.005Z])\}]$; the strongly miscalibrated score is $C[-0.15 + 0.75\,\sigma\{3\,\mathrm{logit}(p(S)) - 0.5\}]$. The simulation seed is 20260411. Figure 5 reports signed bias, coverage of nominal 95% intervals, interval width, and relative efficiency versus PPI++.

We implement two scalar-score versions of RePPI Algorithm 1 (Ji et al., 2025), using three-way sample splitting and weights proportional to the evaluation-fold sizes. RePPI-RF uses a 100-tree random forest. RePPI-MLP-CV selects hidden-layer widths from (8), (32, 32), and (32, 32, 32), and penalties from $\{10^{-2}, 10^{-4}\}$, by two-fold validation MSE within each outer training fold. Both estimate the scalar power-tuning coefficient on the evaluation fold, use the global factor $N/(n+N)$, and set the coefficient to zero when fitted-score variance is numerically singular. The neural implementation uses the feature vector $[1, m(X)]$. Its prediction map and

the scalar single-fold calculation were checked against the public implementation; the three-way aggregation follows the paper's Algorithm 1.

RePPI-RF has lower RMSE than PPI++ in one of 14 settings; RePPI-MLP-CV does so in five. For $n \geq 50$, the neural estimator's RMSE is 0.576–0.668 times that of the labeled mean. At $n = 20$, the ratios are 1.450 and 9.690 under mild and strong miscalibration. Neither RePPI implementation has lower RMSE than smooth monotone calibration in these settings. The largest neural-estimator error occurs when a fitted-score variance of 1.04×10^{-7} produces a coefficient of 1238.2. All replications are retained. Singular-variance fallbacks occur in 175 random-forest replications and 46 neural-network replications; the neural fits produce 2,120 convergence warnings.

This appendix describes the benchmark construction used in section 5.3. We use the official `ppi_py` real-data mean-estimation examples and preserve their labeled-sample-size grids. The `forest` experiment uses $n \in \{50, 100, \ldots, 500\}$. The `galaxies` and semisupervised `census_income` benchmarks use $n \in \{50, 155, 261, 366, 472, 577, 683, 788, 894, 1000\}$. For each n, the remaining observations form the unlabeled sample. We generate 500 random splits for each n, evaluate all estimators on the same split, and use the full-sample mean as the benchmark target.

We compare the labeled-only mean, PPI, AIPW, PPI++, AIPW-EM, `LinearCal`, `MonoSpline`, `IsoCal`, and `AutoCal`. The binary-outcome benchmarks also include imputation, Platt scaling, and Venn–Abers shrinkage. PPI and PPI++ use the official `ppi_py` implementations. PPI++ restricts its coefficient to $[0, 1]$, whereas AIPW-EM minimizes the empirical variance without that constraint.

Calibration maps are fitted on the labeled pairs. We use the calibrated scores in the AIPW estimator, retaining the residual correction after clipping or binning. `LinearCal` fits an affine map; `MonoSpline` fits a smooth nondecreasing spline with a roughness penalty. Both clip predictions to the range of the labeled outcomes. `IsoCal` uses a minimum bin size of 10. Platt scaling fits logistic regression to a stabilized logit transform of the score. `AutoCal` selects among AIPW, linear calibration, monotone splines, and isotonic calibration using 20-fold cross-validation of the estimated influence-function variance. Each fold uses an unlabeled subsample of size $\min(N, 10n)$ to evaluate this criterion. We report Monte Carlo bias, variance, MSE, Wald-interval coverage, and relative efficiency versus PPI.

K.2 LLM score ordering and held-out calibration

For MATH responses evaluated by ArmoRM, empirical correctness increases across twelve score bins, from 0.078 in the lowest bin to 0.920 in the highest. Platt calibration fitted on 100 randomly selected labels reduces mean squared prediction error from 0.4683 to 0.2020 on the remaining held-out observations, a reduction of 56.9%.

K.3 LLM evaluator breakdown

The largest raw-score MSE inflation occurs for Skywork and Athene. With ArmoRM, the main-text estimators have similar MSEs. The evaluator-specific results therefore show where calibration provides the largest gains.

K.4 LLM ranking supplement

The PPE human-preference track also induces a downstream ranking problem across the selected target models. Using the same repeated labeled/unlabeled splits as in section 5.4, we summarize how well each estimator recovers the target-model ordering induced by the full-data human win shares. For each evaluator, labeled-sample size, replicate, and estimator, we rank the eight selected target models by estimated win share, compare that ordering with the ranking induced by the full-data targets, and then macro-average across evaluators. We report Spearman rank correlation, top-1 identification rate, and top-1 regret, defined as the difference in true win share between the best target model and the model selected by the estimator;

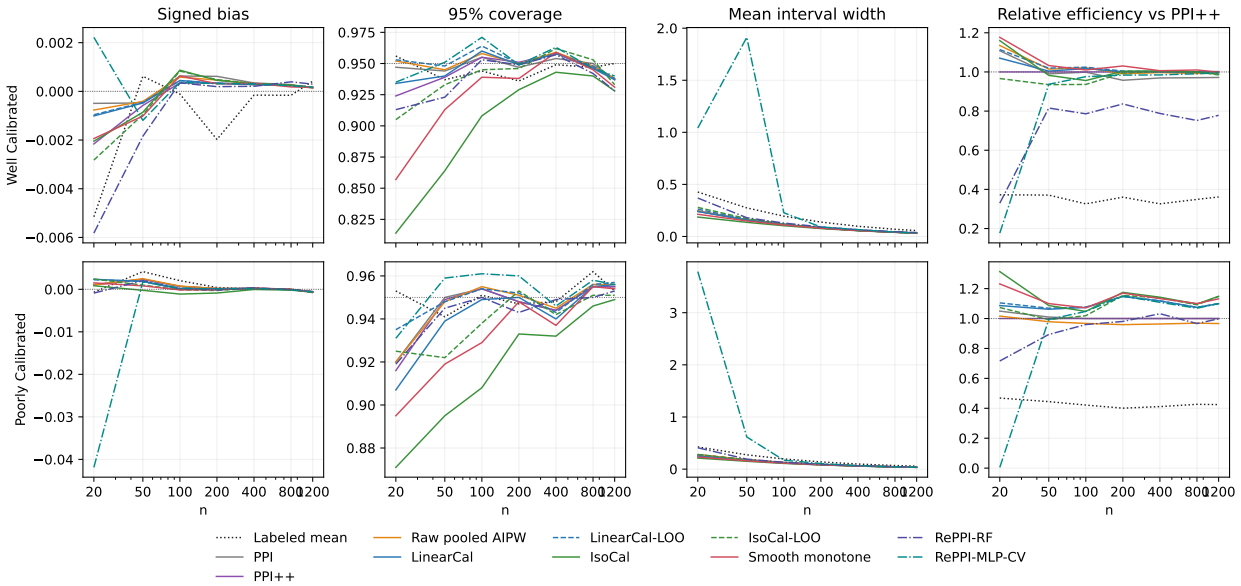


Figure 5: Signed bias, coverage of nominal 95% intervals, interval width, and relative efficiency versus PPI++ in the small-sample experiment. The numerical summary includes Monte Carlo standard errors.

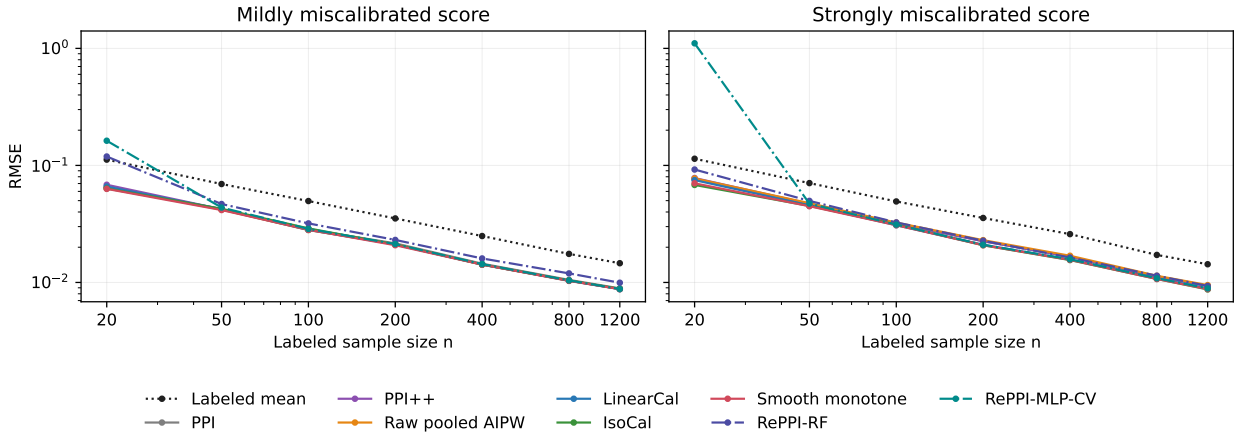


Figure 6: Small-sample RMSE comparison including the random-forest and cross-validated neural-network RePPI implementations.

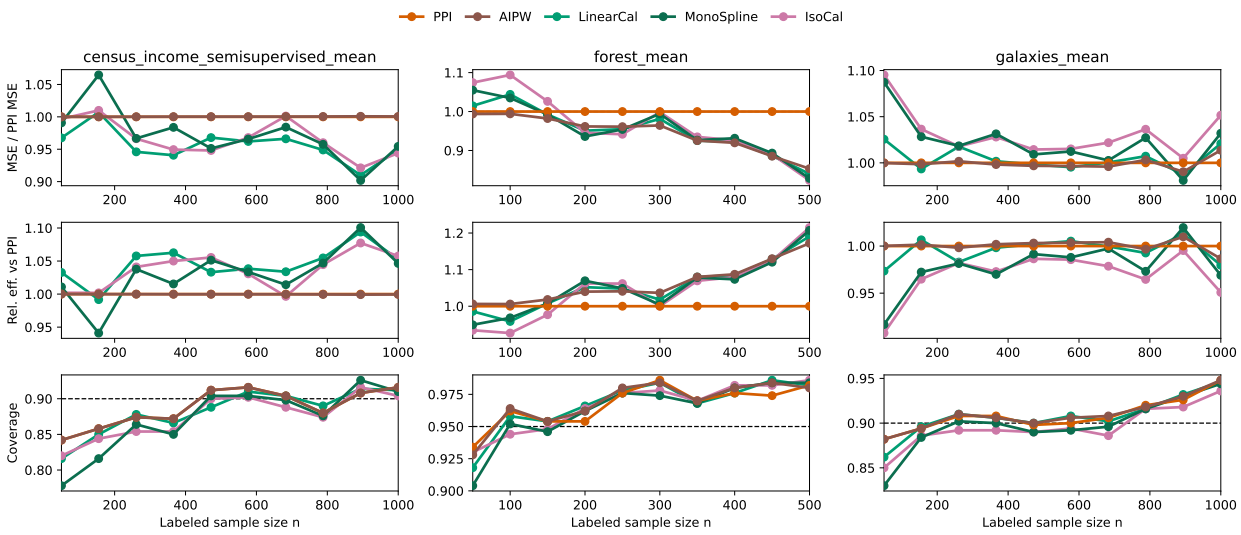


Figure 7: Calibration-focused appendix comparison across the reproduced PPI benchmarks, restricted to PPI, AIPW, `LinearCal`, `MonoSpline`, and `IsoCal`. The panels report normalized MSE relative to PPI, relative efficiency versus PPI, and coverage across labeled-sample-size regimes.

see figure 11 and table 4. The main pattern mirrors the mean-estimation results: calibrated and efficiency-maximized estimators substantially outperform raw PPI and raw AIPW, while remaining close to labeled-only at small n. As shown in figure 11, `AutoCal` attains the highest average Spearman correlation at $n = 200$, about 0.905. By $n = 400$, `LinearCal` gives the best overall ranking quality, with Spearman correlation about 0.941 and top-1 identification rate about 0.646; selected numerical summaries are reported in table 4.

L Code availability

The `ppi_aipw` package, experiment scripts, and numerical results are available in the project repository. Reference implementations are in `examples/paper_calibration.py`. Package documentation includes installation instructions and worked examples.

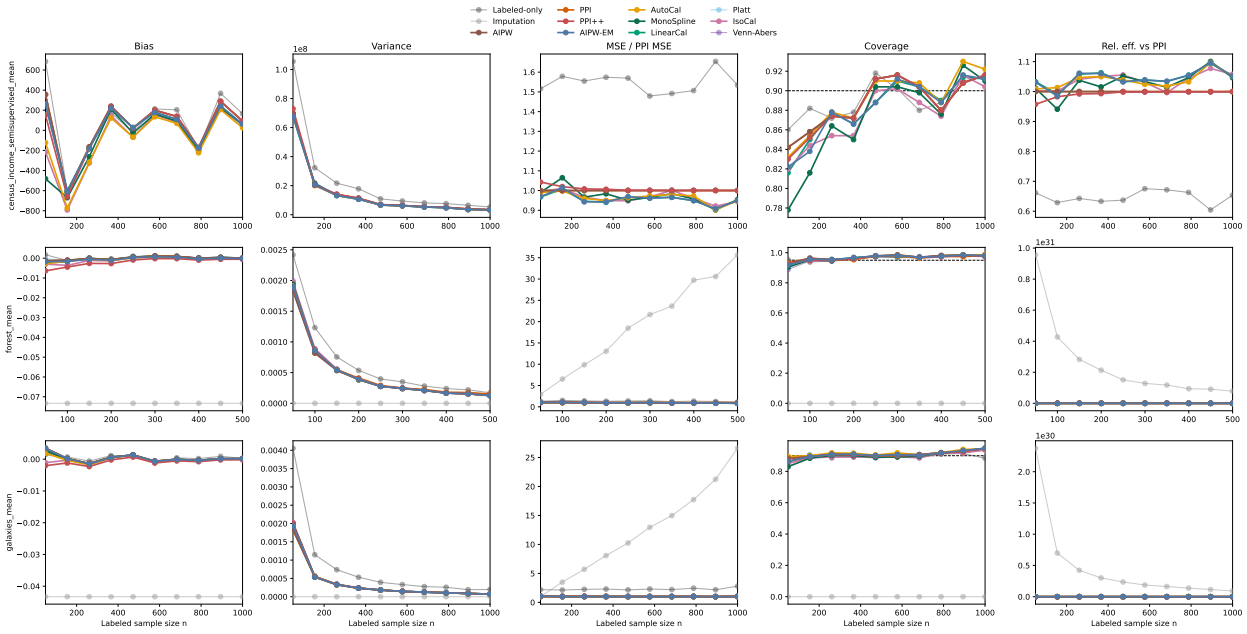


Figure 8: Full diagnostic grid for the reproduced PPI benchmarks, showing bias, empirical variance, normalized MSE relative to PPI, coverage, and relative efficiency versus PPI across labeled-sample-size regimes. In the normalized-MSE and relative-efficiency panels, the dashed horizontal line marks parity with PPI; in the coverage panels it marks the nominal target coverage $1 - \alpha$. `AutoCal`, `MonoSpline`, and `IsoCal` appear alongside PPI, AIPW, PPI++, AIPW-EM, and the fixed calibration rules, while Platt scaling and Venn–Abers appear only on the binary-outcome datasets where they are well defined.

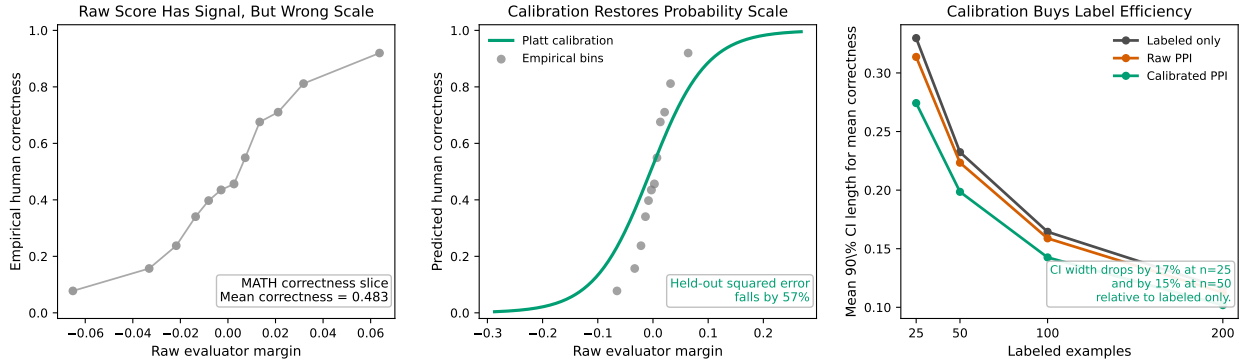


Figure 9: ArmoRM scores and observed MATH correctness. The calibration map is fitted on 100 labeled observations; prediction error is evaluated on the remaining observations.

Dataset	n	N	Estimator	Bias	Variance	MSE	Coverage	RelEff
census_income_semisupervised_mean	50	380041	AIPW	356.4905	69836665.4410	69963750.9366	0.842	1.000
census_income_semisupervised_mean	50	380041	AIPW-EM	260.9833	67728426.6042	67796538.8913	0.822	1.031
census_income_semisupervised_mean	50	380041	AutoCal	-123.2501	69242944.1124	69258134.7078	0.832	1.009
census_income_semisupervised_mean	50	380041	Labeled-only	685.3532	105558402.2427	106028111.3137	0.860	0.662
census_income_semisupervised_mean	50	380041	IsoCal	-215.3832	69697400.2434	69743790.1536	0.820	1.002
census_income_semisupervised_mean	50	380041	LinearCal	260.9294	67622601.4006	67690685.5430	0.816	1.033
census_income_semisupervised_mean	50	380041	MonoSpline	-482.0477	69082916.1270	69315286.1508	0.778	1.011
census_income_semisupervised_mean	50	380041	PPI	356.4473	69834865.9486	69961920.5977	0.842	1.000
census_income_semisupervised_mean	50	380041	PPI++	150.8664	72953053.8328	72975814.4980	0.830	0.957
census_income_semisupervised_mean	577	379514	AIPW	199.8136	6353884.5833	6393810.0648	0.916	1.000
census_income_semisupervised_mean	577	379514	AIPW-EM	177.6067	6114918.2974	6146462.4536	0.912	1.039
census_income_semisupervised_mean	577	379514	AutoCal	134.3746	6201029.6459	6219086.1828	0.910	1.024
census_income_semisupervised_mean	577	379514	Labeled-only	213.2465	9407595.6062	9453069.6946	0.902	0.675
census_income_semisupervised_mean	577	379514	IsoCal	155.0503	6162964.7883	6187005.3977	0.902	1.031
census_income_semisupervised_mean	577	379514	LinearCal	179.8005	6116832.9552	6149161.1594	0.910	1.038
census_income_semisupervised_mean	577	379514	MonoSpline	162.1952	6146069.6864	6172376.9541	0.904	1.033
census_income_semisupervised_mean	577	379514	PPI	199.7932	6351836.6166	6391753.9369	0.916	1.000
census_income_semisupervised_mean	577	379514	PPI++	198.7409	6362047.6345	6401545.5819	0.916	0.998
census_income_semisupervised_mean	1000	379091	AIPW	92.1091	3462051.4638	3470535.5425	0.916	0.999
census_income_semisupervised_mean	1000	379091	AIPW-EM	61.4705	3288235.4312	3292014.0588	0.912	1.052
census_income_semisupervised_mean	1000	379091	AutoCal	25.8207	3289467.9320	3290134.6381	0.922	1.052
census_income_semisupervised_mean	1000	379091	Labeled-only	165.4763	5293296.2822	5320678.6872	0.914	0.654
census_income_semisupervised_mean	1000	379091	IsoCal	51.2697	3271622.4413	3274251.0223	0.904	1.058
census_income_semisupervised_mean	1000	379091	LinearCal	60.7501	3287796.5296	3291487.1058	0.912	1.052
census_income_semisupervised_mean	1000	379091	MonoSpline	60.9626	3306575.6859	3310292.1289	0.910	1.046
census_income_semisupervised_mean	1000	379091	PPI	91.9155	3460071.7259	3468520.1894	0.916	1.000
census_income_semisupervised_mean	1000	379091	PPI++	91.9155	3460071.7259	3468520.1894	0.916	1.000
forest_mean	50	1546	AIPW	-0.0012	0.0018	0.0018	0.928	1.006
forest_mean	50	1546	AIPW-EM	-0.0018	0.0019	0.0019	0.918	0.969
forest_mean	50	1546	AutoCal	-0.0028	0.0019	0.0019	0.924	0.965
forest_mean	50	1546	Labeled-only	0.0017	0.0024	0.0024	0.954	0.762
forest_mean	50	1546	Imputation	-0.0733	0.0000	0.0054	0.000	–
forest_mean	50	1546	IsoCal	-0.0030	0.0020	0.0020	0.930	0.934
forest_mean	50	1546	LinearCal	-0.0016	0.0019	0.0019	0.918	0.986
forest_mean	50	1546	MonoSpline	-0.0021	0.0019	0.0019	0.904	0.949
forest_mean	50	1546	Platt	-0.0003	0.0019	0.0019	0.900	0.948
forest_mean	50	1546	PPI	-0.0013	0.0018	0.0018	0.934	1.000
forest_mean	50	1546	PPI++	-0.0064	0.0019	0.0019	0.928	0.969
forest_mean	50	1546	Venn-Abers	-0.0027	0.0020	0.0020	0.888	0.923
forest_mean	300	1296	AIPW	0.0011	0.0002	0.0002	0.984	1.036
forest_mean	300	1296	AIPW-EM	0.0007	0.0002	0.0002	0.982	1.012
forest_mean	300	1296	AutoCal	0.0006	0.0002	0.0002	0.978	1.002
forest_mean	300	1296	Labeled-only	0.0007	0.0003	0.0004	0.968	0.706
forest_mean	300	1296	Imputation	-0.0733	0.0000	0.0054	0.000	–
forest_mean	300	1296	IsoCal	0.0010	0.0002	0.0002	0.978	1.000
forest_mean	300	1296	LinearCal	0.0010	0.0002	0.0002	0.984	1.018
forest_mean	300	1296	MonoSpline	0.0010	0.0002	0.0002	0.974	1.004
forest_mean	300	1296	Platt	0.0010	0.0002	0.0002	0.982	1.020
forest_mean	300	1296	PPI	0.0011	0.0002	0.0002	0.986	1.000
forest_mean	300	1296	PPI++	-0.0002	0.0002	0.0002	0.980	0.991
forest_mean	300	1296	Venn-Abers	0.0011	0.0002	0.0002	0.978	1.015
forest_mean	500	1096	AIPW	0.0001	0.0001	0.0001	0.980	1.172
forest_mean	500	1096	AIPW-EM	-0.0002	0.0001	0.0001	0.980	1.188
forest_mean	500	1096	AutoCal	-0.0003	0.0001	0.0001	0.986	1.180
forest_mean	500	1096	Labeled-only	0.0001	0.0002	0.0002	0.972	0.879
forest_mean	500	1096	Imputation	-0.0733	0.0000	0.0054	0.000	–
forest_mean	500	1096	IsoCal	-0.0000	0.0001	0.0001	0.986	1.215
forest_mean	500	1096	LinearCal	0.0001	0.0001	0.0001	0.982	1.191
forest_mean	500	1096	MonoSpline	0.0000	0.0001	0.0001	0.984	1.207
forest_mean	500	1096	Platt	0.0001	0.0001	0.0001	0.984	1.203
forest_mean	500	1096	PPI	0.0000	0.0002	0.0002	0.982	1.000
forest_mean	500	1096	PPI++	-0.0006	0.0001	0.0001	0.978	1.179
forest_mean	500	1096	Venn-Abers	-0.0001	0.0001	0.0001	0.978	1.206
galaxies_mean	50	16693	AIPW	0.0035	0.0018	0.0018	0.882	1.000
galaxies_mean	50	16693	AIPW-EM	0.0035	0.0019	0.0020	0.864	0.944
galaxies_mean	50	16693	AutoCal	0.0019	0.0018	0.0018	0.896	0.997
galaxies_mean	50	16693	Labeled-only	0.0014	0.0041	0.0041	0.890	0.451
galaxies_mean	50	16693	Imputation	-0.0433	0.0000	0.0019	0.000	–
galaxies_mean	50	16693	IsoCal	-0.0011	0.0020	0.0020	0.850	0.908
galaxies_mean	50	16693	LinearCal	0.0030	0.0019	0.0019	0.862	0.973
galaxies_mean	50	16693	MonoSpline	0.0026	0.0020	0.0020	0.830	0.917
galaxies_mean	50	16693	Platt	0.0033	0.0019	0.0019	0.856	0.966
galaxies_mean	50	16693	PPI	0.0035	0.0018	0.0018	0.882	1.000
galaxies_mean	50	16693	PPI++	-0.0020	0.0020	0.0020	0.864	0.914
galaxies_mean	50	16693	Venn-Abers	0.0022	0.0020	0.0020	0.830	0.907
galaxies_mean	577	16166	AIPW	-0.0006	0.0001	0.0001	0.906	1.004
galaxies_mean	577	16166	AIPW-EM	-0.0007	0.0001	0.0001	0.906	1.005
galaxies_mean	577	16166	AutoCal	-0.0008	0.0001	0.0001	0.918	0.989
galaxies_mean	577	16166	Labeled-only	-0.0011	0.0003	0.0003	0.910	0.436
galaxies_mean	577	16166	Imputation	-0.0433	0.0000	0.0019	0.000	–
galaxies_mean	577	16166	IsoCal	-0.0007	0.0001	0.0001	0.894	0.986
galaxies_mean	577	16166	LinearCal	-0.0007	0.0001	0.0001	0.908	1.005
galaxies_mean	577	16166	MonoSpline	-0.0006	0.0001	0.0001	0.892	0.988
galaxies_mean	577	16166	Platt	-0.0007	0.0001	0.0001	0.896	1.002
galaxies_mean	577	16166	PPI	-0.0006	0.0001	0.0001	0.900	1.000
galaxies_mean	577	16166	PPI++	-0.0012	0.0001	0.0001	0.902	0.988
galaxies_mean	577	16166	Venn-Abers	-0.0006	0.0001	0.0001	0.892	0.986
galaxies_mean	1000	15743	AIPW	0.0002	0.0001	0.0001	0.948	0.987
galaxies_mean	1000	15743	AIPW-EM	0.0002	0.0001	0.0001	0.948	0.980
galaxies_mean	1000	15743	AutoCal	0.0001	0.0001	0.0001	0.944	0.977
galaxies_mean	1000	15743	Labeled-only	0.0003	0.0002	0.0002	0.884	0.357
galaxies_mean	1000	15743	Imputation	-0.0433	0.0000	0.0019	0.000	–
galaxies_mean	1000	15743	IsoCal	0.0001	0.0001	0.0001	0.936	0.951
galaxies_mean	1000	15743	LinearCal	0.0002	0.0001	0.0001	0.946	0.980
galaxies_mean	1000	15743	MonoSpline	0.0002	0.0001	0.0001	0.944	0.969
galaxies_mean	1000	15743	Platt	0.0002	0.0001	0.0001	0.942	0.979
galaxies_mean	1000	15743	PPI	0.0002	0.0001	0.0001	0.948	1.000
galaxies_mean	1000	15743	PPI++	-0.0002	0.0001	0.0001	0.946	0.977
galaxies_mean	1000	15743	Venn-Abers	0.0002	0.0001	0.0001	0.942	0.952

Table 2: Representative numerical summary at the smallest, middle, and largest labeled sample sizes for each reproduced benchmark. The reported quantities are Monte Carlo bias, empirical variance of the point estimator, MSE, Wald-interval coverage, and relative efficiency versus PPI for the displayed benchmark set, including PPI++, AIPW-EM, `AutoCal`, `MonoSpline`, and `IsoCal`. Relative efficiency is not reported when the estimator has zero variance.

Track	n	Estimator	MSE / PPI	Label savings	Coverage
PPE Correctness	100	Labeled-only	0.346	0.000	0.927
PPE Correctness	100	PPI	1.000	0.010	0.904
PPE Correctness	100	AIPW	0.774	0.009	0.906
PPE Correctness	100	PPI++	0.330	0.070	0.929
PPE Correctness	100	AIPW-EM	0.309	0.095	0.930
PPE Correctness	100	LinearCal	0.305	0.101	0.931
PPE Correctness	100	AutoCal	0.310	0.094	0.929
PPE Correctness	100	MonoSpline	0.309	0.092	0.926
PPE Correctness	100	IsoCal	0.309	0.089	0.925
PPE Correctness	100	Platt	0.304	0.102	0.930
PPE Correctness	400	Labeled-only	0.299	0.000	0.988
PPE Correctness	400	PPI	1.000	0.020	0.924
PPE Correctness	400	AIPW	0.405	0.009	0.941
PPE Correctness	400	PPI++	0.276	0.083	0.989
PPE Correctness	400	AIPW-EM	0.262	0.103	0.989
PPE Correctness	400	LinearCal	0.260	0.105	0.989
PPE Correctness	400	AutoCal	0.263	0.102	0.988
PPE Correctness	400	MonoSpline	0.263	0.094	0.989
PPE Correctness	400	IsoCal	0.265	0.089	0.989
PPE Correctness	400	Platt	0.260	0.105	0.989
PPE Human Preference	100	Labeled-only	0.303	0.000	0.911
PPE Human Preference	100	PPI	1.000	0.004	0.897
PPE Human Preference	100	AIPW	0.919	0.004	0.896
PPE Human Preference	100	PPI++	0.298	0.027	0.912
PPE Human Preference	100	AIPW-EM	0.295	0.031	0.909
PPE Human Preference	100	LinearCal	0.295	0.031	0.909
PPE Human Preference	100	AutoCal	0.297	0.028	0.908
PPE Human Preference	100	MonoSpline	0.297	0.025	0.903
PPE Human Preference	100	IsoCal	0.297	0.024	0.905
PPE Human Preference	400	Labeled-only	0.289	0.000	0.945
PPE Human Preference	400	PPI	1.000	0.005	0.915
PPE Human Preference	400	AIPW	0.703	0.004	0.917
PPE Human Preference	400	PPI++	0.283	0.025	0.946
PPE Human Preference	400	AIPW-EM	0.282	0.028	0.945
PPE Human Preference	400	LinearCal	0.282	0.028	0.944
PPE Human Preference	400	AutoCal	0.282	0.027	0.944
PPE Human Preference	400	MonoSpline	0.282	0.027	0.942
PPE Human Preference	400	IsoCal	0.282	0.026	0.940

Table 3: PPE numerical summary at $n \in \{100, 400\}$, including the efficiency-maximized and additional fixed-calibration estimators.

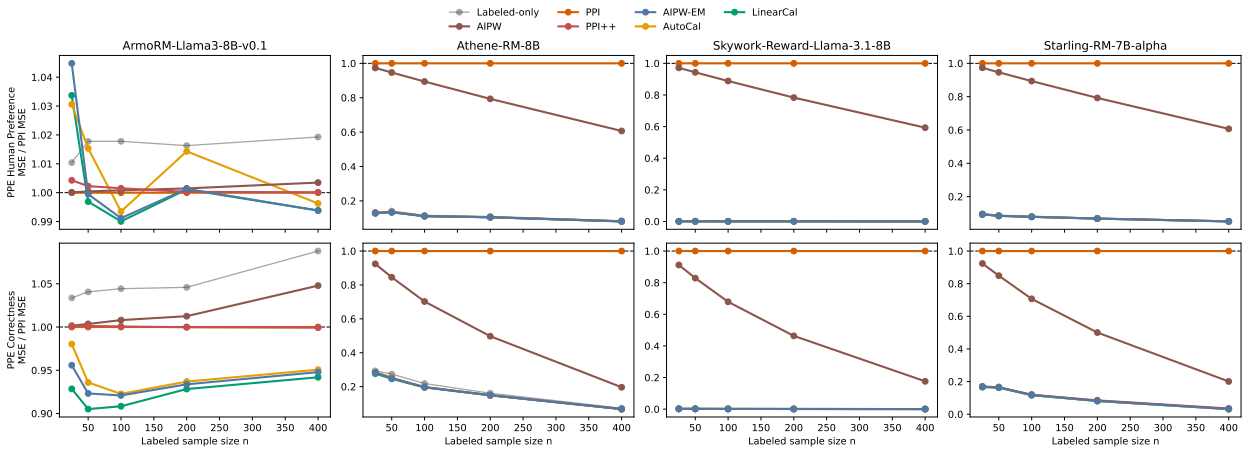


Figure 10: Appendix evaluator-specific PPE summary. Each panel repeats the PPE Human and PPE Correctness comparisons for one public evaluator. The largest MSE increases relative to labeled-only estimation occur for Skywork and Athene.

n	Estimator	Spearman	Top-1	Top-1 regret
25	AIPW	0.268	0.216	0.064
25	AIPW-EM	0.599	0.344	0.028
25	AutoCal	0.600	0.343	0.028
25	Labeled-only	0.606	0.315	0.030
25	LinearCal	0.599	0.344	0.028
25	PPI	0.265	0.216	0.064
25	PPI++	0.594	0.336	0.029
100	AIPW	0.468	0.285	0.044
100	AIPW-EM	0.845	0.477	0.007
100	AutoCal	0.845	0.487	0.007
100	Labeled-only	0.846	0.485	0.007
100	LinearCal	0.846	0.482	0.007
100	PPI	0.456	0.279	0.046
100	PPI++	0.844	0.479	0.007
400	AIPW	0.631	0.439	0.027
400	AIPW-EM	0.941	0.644	0.002
400	AutoCal	0.941	0.640	0.002
400	Labeled-only	0.940	0.625	0.002
400	LinearCal	0.941	0.646	0.002
400	PPI	0.579	0.419	0.031
400	PPI++	0.940	0.636	0.002

Table 4: Appendix PPE-ranking summary at selected labeled sample sizes from the LLM benchmark. Better rankings have higher Spearman correlation and top-1 identification rates and lower top-1 regret.

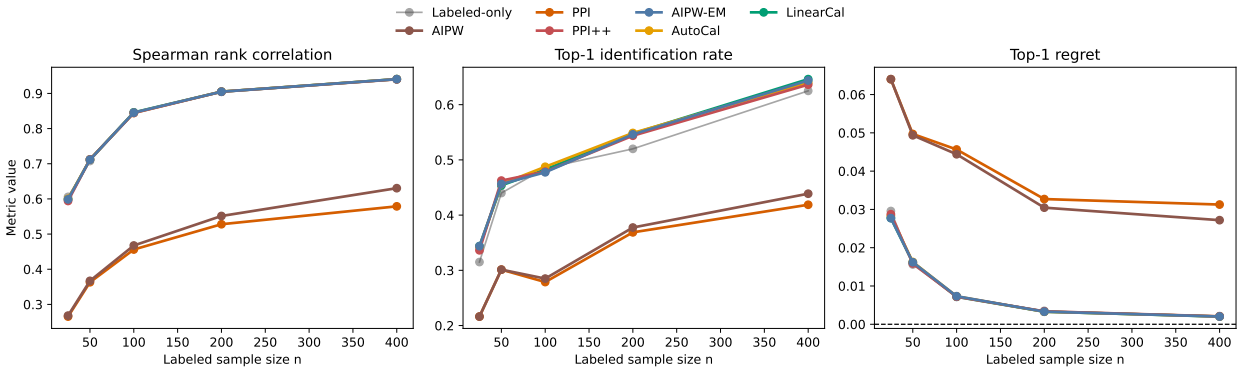


Figure 11: Appendix PPE-ranking supplement from the LLM benchmark, macro-averaged across the public evaluator models. The three panels report Spearman rank correlation, top-1 identification rate, and top-1 regret for recovering the ranking of the eight selected PPE target models from a small labeled sample plus evaluator scores.